



Review article

Decoding senescent drivers in Alzheimer's disease: From bench to bedside

Runjin Zhou^a, Xiaoling Lin^b, Zhenge Liao^c, Jian Lin^d, Ruixue Zheng^c, Haojie Ni^a, Wei Liu^c, Zhanpeng Feng^c, Qiuxing He^a, Xingdong Lin^{e,*}, Chunzhi Tang^{c,*}, Juxian Song^{f,g,**}, Weimin Ning^{a,h,i,***}

^a Dongguan Hospital of Traditional Chinese Medicine, Dongguan, China

^b Guangzhou Women and Children's Medical Center, Guangzhou, China

^c Medical College of Acupuncture-Moxibustion and Rehabilitation, Guangzhou University of Chinese Medicine, Guangzhou, China

^d The Third Clinical Medical College, Guangzhou University of Chinese Medicine, Guangzhou, China

^e The Third Affiliated Hospital, Guangzhou University of Chinese Medicine, Guangzhou, China

^f School of Pharmacy, Chengdu University of Traditional Chinese Medicine, Chengdu, China

^g Mr. & Mrs. Ko Chi-Ming Centre for Parkinson's Disease Research (CPDR), School of Chinese Medicine, Hong Kong Baptist University, Hong Kong SAR, China

^h Dongguan Key Laboratory of Intractable Brain Diseases in Dongguan, Dongguan Hospital of Guangzhou University of Chinese Medicine, Dongguan, China

ⁱ State Key Laboratory of Dampness Syndrome of Chinese Medicine, Dongguan Hospital of Guangzhou University of Chinese Medicine, Dongguan, China

ARTICLE INFO

Keywords:

Alzheimer's disease
cellular senescence
senolytics
senomorphics

ABSTRACT

Alzheimer's disease (AD) is the most common neurodegenerative disorder associated with dementia. Cellular senescence, widely acknowledged as a key hallmark of aging, has increasingly been recognized as a significant factor in the pathogenesis of AD, although the precise mechanisms underlying this relationship have yet to be fully understood. In the brains of individuals with AD, neurons, glial cells, and cerebrovascular endothelial cells exhibit premature senescence, characterized by irreversible cell cycle arrest, resistance to apoptosis, and the secretion of a diverse range of bioactive molecules collectively referred to as the senescence-associated secretory phenotype (SASP). These senescent cells profoundly influence the neural microenvironment through the release of SASP factors, thus exacerbating A β - and tau-induced neurotoxicity, promoting neuroinflammatory responses, and impairing the integrity of the blood-brain barrier (BBB), ultimately giving rise to a self-sustaining "senescence-neurodegeneration" cycle. Despite progress in therapies targeting A β and tau pathology, their clinical effectiveness remains limited, highlighting the urgent need for alternative therapeutic strategies. This review presents a comprehensive analysis of the molecular mechanisms connecting AD with cellular senescence, examines how the senescent microenvironment contributes to neurodegeneration, and evaluates the therapeutic potential of senotherapeutic interventions—including senolytics and senomorphics—as novel approaches for the clinical management of AD.

1. Introduction

Alzheimer's disease (AD), as the most common neurodegenerative disease and the primary cause of dementia, has witnessed a steady and marked rise in its global disease burden. According to the 2023 annual

report released by the Alzheimer's Disease International (ADI), there are currently approximately 55 million people with dementia worldwide, with AD accounting for about 60–70 %. It is projected to rise to 78 million by 2030 and surge sharply to 139 million by 2050 (Long et al., 2023). Under the traditional framework constructed by core

Abbreviations: A β , Amyloid-beta; AD, Alzheimer's disease; ATM, Ataxia telangiectasia mutated; ATR, Ataxia telangiectasia and Rad3-related protein; BBB, Blood-brain barrier; DDR, DNA damage response; DSBs, double-strand breaks; HMGB1, High mobility group box 1 Protein; NFTs, Neurofibrillary tangles; OPCs, Oligodendrocyte progenitor cells; OXPHOS, Oxidative phosphorylation; RB, Retinoblastoma; ROS, Reactive oxygen species; NSCs, Neural stem cells; SA- β -gal, SA- β -galactosidase; SAHF, Senescence-associated heterochromatin foci; SASP, Senescence-associated secretory phenotype; SCAP, Senescent cell anti-apoptotic pathway; SnCs, Senescent cells.

* Corresponding authors.

** Corresponding author at: School of Pharmacy, Chengdu University of Traditional Chinese Medicine, Chengdu, China.

*** Corresponding author at: Dongguan Hospital of Traditional Chinese Medicine, Dongguan, China.

E-mail addresses: linxdlili@gzucm.edu.cn (X. Lin), jordan664@163.com (C. Tang), juxian.song@gmail.com (J. Song), naobingke36@163.com (W. Ning).

<https://doi.org/10.1016/j.arr.2025.102978>

Received 7 August 2025; Received in revised form 26 November 2025; Accepted 28 November 2025

Available online 2 December 2025

1568-1637/© 2025 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

mechanisms such as the amyloid β -protein (A β)/tau cascade, neuroinflammation, and vascular damage, the therapeutic bottleneck of AD has become increasingly prominent. Monoclonal antibody drugs targeting A β , including Aducanumab, Lecanemab, and Donanemab, have been approved by the FDA between 2021 and 2024, showing potential in clearing A β plaques and slowing cognitive decline in AD. Phase III trials showed that they can clear amyloid plaques and delay cognitive decline in early-stage (Sims et al., 2023; van Dyck et al., 2023), but they are only applicable to patients with mild symptoms, and are associated with risks of cerebral edema and microhemorrhage (Daly et al., 2024; Rabinovici, 2021; van Dyck et al., 2023). Their high cost also limits their clinical application. Emerging tau-targeted therapies have entered Phase II clinical trials, but no breakthrough in efficacy has been achieved yet (Edwards et al., 2023; Grüniger, 2015; Mummery et al., 2023).

These findings have prompted researchers to delve into the deeper biological drivers in the pathological process. Cellular senescence, a hallmark of aging, has emerged as a pivotal driver and synergistic factor in the progression of AD. It influences the pathological landscape of AD by altering interactions between various cell types, thereby reshaping the disease's microenvironment. The concept of cellular senescence was first proposed by Hayflick and Moorhead in 1961. Through in vitro experiments, they demonstrated that human primary fibroblasts reach a finite limit of proliferative potential during continuous subculture. It is manifested as a state where cells enter an irreversible, non-dividing state with permanent cell cycle arrest, yet remain viable (Hayflick, 1965). The concept of "replicative senescence" was thus put forward, which is most

likely caused by the gradual shortening of telomeres in proliferating cells (de Lange, 1998; Harley et al., 1990). In the absence of telomerase, telomeres shorten progressively as cells continue to divide, thereby triggering the activation of senescence signals (Harley et al., 1990; Kiyono et al., 1998; Vaziri and Benchimol, 1998). In addition to "replicative senescence", there is also "stress-induced senescence", which refers to the occurrence of cell cycle arrest caused by DNA damage from ionizing radiation, mitochondrial dysfunction, abnormal oncogene activation, oxidative stress, and impaired autophagy (Herranz and Gil, 2018) (Fig. 1).

As cellular senescence has been established as a strong risk factor for AD, this biological state—characterized by cell cycle arrest, resistance to apoptosis, and the senescence-associated secretory phenotype (SASP)—has gradually moved from the margins of aging research to the center of neurodegenerative disease exploration. Recent breakthrough studies have revealed that neurons, glial cells, and cerebrovascular endothelial cells in the brains of AD patients generally exhibit an accelerated senescence phenotype. These cells actively reshape the neural microenvironment by releasing SASP factors. This process directly amplifies the neurotoxicity of A β and tau, exacerbates neuroinflammatory storms, impairs the integrity of the blood-brain barrier (BBB), and forms a cross-pathological "senescence-neurodegeneration" vicious cycle (Gaikwad et al., 2021; Zhang et al., 2019). More crucially, the clearance of senescent cells (SnCs) can significantly alleviate core pathologies in AD, and mitigate cognitive decline (Millar et al., 2025). Preclinical evidence further supports this, showing that selective clearance of senescent glial cells in PS19 tau transgenic mice reduces tau phosphorylation and

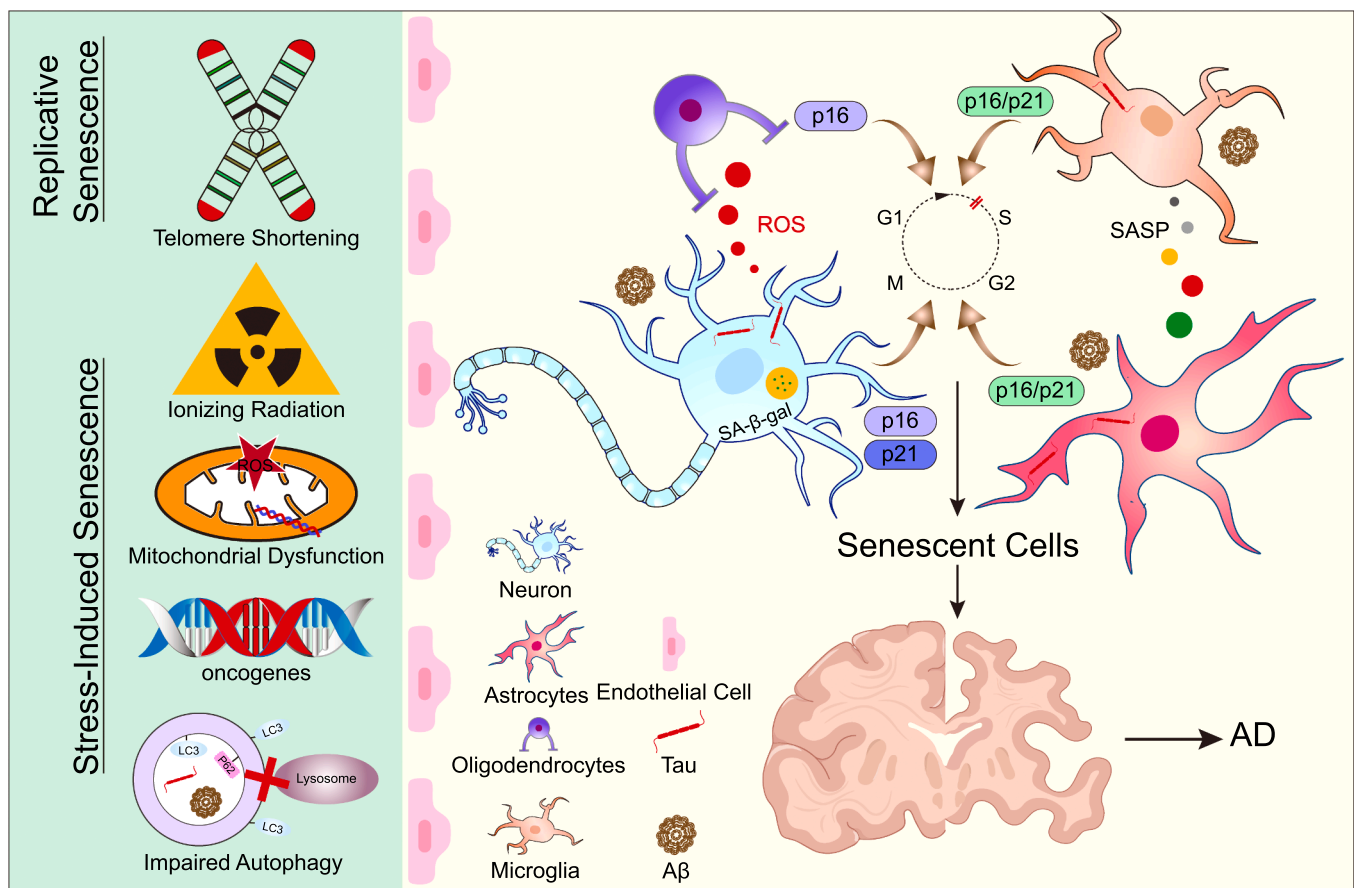


Fig. 1. Mechanistic Links Between Cellular Senescence and Alzheimer's disease (AD). The diagram illustrates two primary senescence induction mechanisms: replicative senescence (triggered by telomere shortening) and stress-induced senescence (driven by ionizing radiation, mitochondrial dysfunction, oncogene activation, or impaired autophagy). Senescent cells (SnCs) exhibit hallmark markers (p16, p21) and generate a senescence-associated secretory phenotype (SASP), accompanied by reactive oxygen species (ROS) production. Multiple central nervous system cell types, including neurons, microglia, oligodendrocytes, and astrocytes, are implicated in propagating senescence-associated pathology. Key disease-associated proteins (tau, A β) are linked to major AD.

neurofibrillary tangles (NFTs), while restoring cognitive function (Bussian et al., 2018). Collectively, these pieces of evidence identify cellular senescence as a pivotal node in the pathogenesis of AD. It not only serves as a key biological bridge linking the normal aging process to neurodegenerative lesions but also acts as an integrated regulatory network that coordinates multiple pathological processes, such as the A β /tau pathological cascade, neuroinflammatory storms, and vascular dysfunction.

However, the current knowledge system in this field remains highly fragmented. Important knowledge gaps that have not yet been bridged and urgently require systematic analysis include the spatiotemporal heterogeneity of the dynamic distribution of SnCs in the brain, the differential damage mechanisms of SASP factors on specific neural circuits, and the specific response patterns of senolytics therapy in different cell types. This is precisely the core driving force behind this review: to

systematically sort out the molecular networks of the interaction between AD and cellular senescence, analyze how the senescent micro-environment drives neurodegeneration, and critically evaluate the translational prospects and existing challenges of intervention strategies such as senolytics and senomorphics, thereby providing a new perspective for overcoming AD.

2. Cell-cycle arrest pathways (p53/p21 and p16/RB) driving cellular senescence in AD

During the progression of AD, abnormal activation of the p53/p21 and p16/RB signaling pathways constitutes the core molecular hub driving senescence in neurons and glial cells. These pathways trigger irreversible cell cycle arrest in response to various AD-related stressors (Nakamura et al., 2008; Rodier and Campisi, 2011). Specifically, the

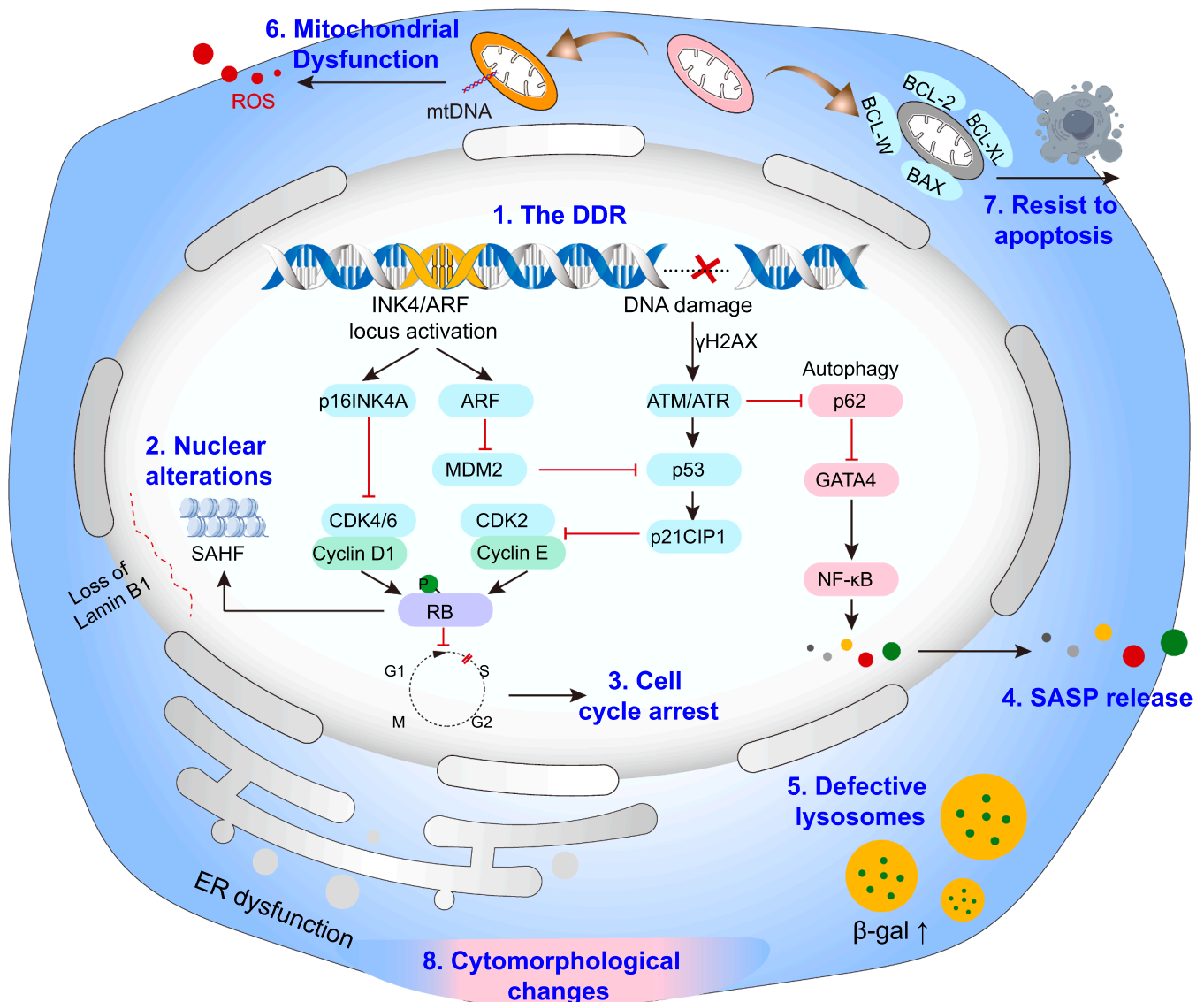


Fig. 2. Schematic illustration of the core interconnected molecular mechanisms underlying cellular senescence in Alzheimer's disease (AD) pathogenesis, encompassing the key hallmarks of this process. The process initiates with the activation of the DNA damage response (DDR, 1), which triggers nuclear alterations (2), including activation of the INK4/ARF locus, loss of Lamin B1, and formation of senescence-associated heterochromatin foci (SAHF). This subsequently leads to cell cycle arrest (3) through the p16/RB and p53/p21 signaling pathways. Downstream events include the release of the senescence-associated secretory phenotype (SASP) (4), mediated by GATA4/NF-κB activation; lysosomal dysfunction characterized by increased β-galactosidase (β-gal) activity (5) and impaired autophagy; mitochondrial dysfunction resulting in the generation of reactive oxygen species (ROS, 6); resistance to apoptosis regulated by BCL-2 family proteins (7); endoplasmic reticulum (ER) stress; and cytomorphological changes (8). Arrows indicate key signaling interactions and pathway dependencies. Collectively, these elements may contribute to the progression of cellular senescence in the context of AD development.

p53/p21 pathway is primarily activated by DNA double-strand breaks (DSBs): Reactive oxygen species (ROS) bursts induced by A β oligomers and telomere shortening continuously cause genomic damage, activating the ataxia telangiectasia mutated (ATM) gene and ataxia telangiectasia and Rad3-related protein (ATR). Subsequently, ATM and ATR can activate checkpoint kinases CHK1 and CHK2 (Buscemi et al., 2004; d'Adda di Fagagna, 2008), promoting the phosphorylation of serine 15 (Ser15) in the p53 protein, stabilizing its conformation, and enhancing its transcriptional activity (Smits et al., 2006). Nuclear p53 directly binds to the p21 promoter region, significantly upregulating p21 expression. As a cyclin-dependent kinase inhibitor, p21 prevents the phosphorylation of retinoblastoma (RB) protein by inhibiting the activity of the cyclin E-CDK2 complex, leading to the inactivation of E2F transcription factors and ultimately arresting cells in the G1/S phase (Herranz and Gil, 2018; Martínez-Zamudio et al., 2017) (Fig. 2).

Another senescence activation mechanism involves the de-repression of the INK4-ARF gene. In normal cells, the INK4-ARF locus is repressed by polycomb proteins and epigenetic factors, but upon activation, it promotes the expression of ARF and p16. As a specific CDK4/6 inhibitor, p16 prevents the phosphorylation of RB protein by the cyclin D-CDK4/6 complex, maintaining RB in an active state. By directly binding to and inhibiting E2F-dependent gene transcription, p16 forces cells to arrest in the G1 phase (Saez-Atienzar and Masliah, 2020) (Fig. 2). In terms of spatial distribution, p16-positive senescent cells are highly enriched in the hippocampal CA1 region and prefrontal cortex of AD patients, and co-localize with phosphorylated tau (Graves et al., 2025).

Crucially, these two pathways exhibit synergistic interaction in AD: On one hand, p21 can indirectly enhance the hypophosphorylated state of RB by inhibiting CDK2, amplifying the senescence effect of the p16/RB pathway. On the other hand, p53 activated by persistent oxidative stress can directly upregulate p16 expression. This cross-regulation leads to a "senescence lock" effect, meaning that senescent cells cannot re-enter the cell cycle even after being separated from the initial stressor (Rodier et al., 2009). Moreover, p16 induces endothelial extracellular matrix (ECM) dysregulation in PS19 tau mice, and its knockout ameliorates this pathological alteration; the p16-ECM axis is implicated in tau-dependent neurodegeneration (Graves et al., 2025). In summary, the p53/p21 and p16/RB pathways not only constitute the molecular engine of cellular senescence in AD, but the DNA damage-cycle arrest-inflammation cascade they mediate is also directly involved in the neurodegenerative process, making them key targets linking aging biology to multiple pathological dimensions of AD (Ahmad et al., 2024).

3. Characteristics of cellular senescence

In the brains of AD patients, cellular senescence is often accompanied by the following characteristics (Suryadevara et al., 2024): permanent cell cycle arrest, DNA damage, epigenetic changes, upregulation of anti-apoptotic factors, production of SASP, increased β -galactosidase (β -gal) activity, changes in mitochondrial energy metabolism, and cytomorphological changes (Fig. 2). Among these, permanent cell cycle arrest and changes in mitochondrial function are regarded as hallmarks of cellular senescence, while the production of SASP and increased β -gal activity are typical features of cellular senescence (Si et al., 2021).

3.1. The DNA damage response

In the brains of patients with AD, the DNA damage response (DDR) pathways in neurons and glial cells exhibit a state of persistent activation. This activation is primarily attributed to the significant accumulation of DNA damage, particularly oxidative stress-induced DSBs and base damage represented by 8-hydroxy-2'-deoxyguanosine (8-OHdG), which is marked by the persistent aggregation of phosphorylated histone H2AX (γ H2AX) and p53-binding protein 1 (53BP1) at damage foci (Shanbhag et al., 2019). Such damage is driven by multiple AD-specific factors: Firstly, oxidative stress induced by A β oligomers directly attacks

DNA (Suberbielle et al., 2013). Secondly, age-related telomere shortening and epigenomic instability further impair the cell's repair capacity. Additionally, the decreased function of the key repair enzyme PARP1 in senescent cells collectively forms a vicious cycle (Kolacz and Robaszkiewicz, 2024). Sustained DDR leads to abnormalities and chronic activation of the ATM/ATR-CHK1/CHK2 signaling pathway, which in turn drives permanent cell cycle arrest, ultimately inducing cellular senescence through the p53/p21 or p16/RB pathways.

3.2. Nuclear alterations

During the neuronal senescence process in AD, the nuclear structure undergoes systematic disruption, manifested as nuclear membrane invagination, abnormalities of nuclear pore complexes, significant loss of Lamin B1, and deficiency of heterochromatin (senescence-associated heterochromatin foci, SAHF) (Eftekharpour, 2023). As a key protein maintaining nuclear membrane integrity and heterochromatin organization, the downregulation of Lamin B1 directly triggers a cascade of pathological effects, including SAHF formation: Firstly, it leads to the disordered distribution of heterochromatin-associated protein HP1 α and the de-repression of satellite DNA repeat sequences, promoting chromatin structure relaxation, abnormal activation of transposons, and exacerbating genomic instability (Swanson et al., 2013, 2015). Secondly, the disruption of nuclear membrane structure and nuclear pore dysfunction are further aggravated by abnormal phosphorylation of tau protein, which, by impeding nucleocytoplasmic transport pathways, induces the mislocalization of RNA-binding proteins such as TDP-43 to the cytoplasm and the formation of pathological inclusion bodies (Frost et al., 2016). Meanwhile, abnormal nucleolar morphology in AD neurons indicates severe nucleolar stress, causing repression of rDNA transcription and impairment of ribosome biogenesis, directly weakening the ability to maintain protein homeostasis (Kiryk et al., 2013). Notably, NFTs formed by misfolded tau protein in the nucleus, as typical inclusion bodies, not only directly disrupt gene transcriptional regulation but also synergize with the aforementioned nuclear structural lesions to drive neuronal dysfunction, ultimately leading to cognitive decline (Maina et al., 2018).

3.3. Resist to apoptosis

A key survival feature of senescent cells is their high resistance to apoptotic signaling pathways. In senescent neurons within the AD brain, this anti-apoptotic property is manifested by the upregulation of anti-apoptotic BCL-2 protein family members (BCL-2 and BCL-XL) and the inhibition of the activity of pro-apoptotic members (BAX and BAK), forming the senescent cell anti-apoptotic pathway (SCAP) (Childs et al., 2014; Yosef et al., 2016). Key regulatory factors include the senescence-specifically overexpressed p53 isoform Δ 133p53, which inhibits the pro-apoptotic activity of full-length p53 (Gong et al., 2015). Meanwhile, the expression of pro-apoptotic proteins PUMA and NOXA is also inhibited or their functions are antagonized (Tavana et al., 2010). Cellular senescence is accompanied by the stable expression of the endogenous inhibitor of apoptosis proteins (IAPs) family, which includes XIAP and cIAP1 and further reinforces apoptotic resistance (Hu et al., 2022). This "zombie" state allows senescent nerve cells to survive for a long time, but the harmful SASP factors they secrete continuously damage the surrounding microenvironment.

3.4. SASP release

SASP, the most prominent effector characteristic of senescent cells and a primary driver of AD progression, refers to the secretion of a large number of soluble factors with pro-inflammatory, pro-apoptotic, profibrotic, and pro-tumorigenic activities. In the AD brain, senescent neurons and glia release a rich array of SASP, including pro-inflammatory cytokines (IL-1 α , IL-1 β , IL-6, TNF- α), chemokines

(CXCL1, CXCL8, MCP-1), growth factors (VEGF, TGF- β , HGF), matrix metalloproteinases and their inhibitors (MMP-3, MMP-9, PAI-1), ROS/nitrogen species (RNS), as well as damage-associated molecular patterns (such as high mobility group box 1 Protein, HMGB1) (Coppé et al., 2008).

Studies have demonstrated that SASP components can regulate various pathophysiological effects of senescent cells through paracrine and autocrine mechanisms (Malaquin et al., 2016). In terms of paracrine action, SASP factors, such as IL-6 and INF γ from astrocyte, are released into the brain microenvironment, where they bind to and activate IL-6R receptor on adjacent normal neurons and microglia, thereby inducing a senescent phenotype in these cells (secondary senescence). This paracrine-mediated amplification drives a self-perpetuating pathological cycle that accelerates the progression of AD (Han et al., 2020). In autocrine action, SASP factors directly act on the NOTCH1-JAG1 receptors on the surface of the senescent cells that secrete them (Teo et al., 2019). On one hand, they continuously activate pro-inflammatory and pro-survival pathways such as NF- κ B in senescent cells, strengthening and maintaining their senescent phenotype. On the other hand, they inhibit key senescence-inhibiting pathways, hindering the clearance of senescent cells. During the progression of AD, these factors play a core role: IL-1 β and TNF- α can directly impair neuronal synaptic plasticity and enhance A β production (Carlos et al., 2017; Terrando et al., 2010); TGF- β stimulates the activation of astrocytes (Preininger et al., 2023); MMPs damage the integrity of the BBB and abnormally process amyloid precursor protein (APP) (Hernandez-Guillamon et al., 2015); while ROS and HMGB1 synergistically activate neurotoxic reactions mediated by astrocyte (Gaikwad et al., 2021). The expression of SASP is mainly driven by multiple signaling pathways including NF- κ B, C/EBP β , IL-1 α autocrine positive feedback, and p38MAPK (Wang et al., 2024).

3.5. Defective lysosomes

Lysosomal dysfunction is a hallmark of cellular senescence in AD, characterized by lysosomal membrane permeabilization (LMP), reduced enzymatic activity, and autophagic flux arrest. The accumulation of giant lysosomes is commonly observed in AD neurons, which are rich in undigested lipofuscin—a terminal product of autophagic degradation composed of oxidized lipids, proteins, and metals (Whyte et al., 2017). A significant impairment in the autophagy-lysosomal pathway is a signature molecular event in AD. For one thing, presenilin 1 (PSEN1) mutations impair the activity of the proton pump v-ATPase, leading to an increase in intralysosomal pH and a reduction in the activity of cathepsins B and D (CTSB/D) (Lee et al., 2010). For another, A β oligomers and tau aggregates themselves can disrupt the stability of the lysosomal membrane. Additionally, autophagic flux is blocked in AD neurons, with impaired fusion between autophagosomes and lysosomes (defective TFEB nuclear translocation) and impaired lysosomal biogenesis (mTORC1 inhibition), collectively resulting in a significant decrease in the clearance rate of misfolded protein aggregates (Abokyi et al., 2023). Ultimately, lysosomal dysfunction results in the accumulation of metabolic waste products and induces an energy crisis in senescent cells, including neurons and glial cells, the latter of which critically depend on lysosomal activity for the phagocytic clearance of A β and apoptotic debris (Sun et al., 2025; Tan and Finkel, 2023). Lysosomal stress caused by the long-term accumulation of pathological products triggers the activation of the NLRP3 inflammasome, further releasing inflammatory factors such as IL-1 β and forming a positive feedback loop of neuroinflammation.

3.6. Mitochondrial dysfunction

In senescent neurons of AD, mitochondrial homeostasis is disrupted, presenting multiple abnormalities in structure and function (D'Alessandro et al., 2025; Wang et al., 2020): Structurally, mitochondria undergo fragmentation, with disordered or broken cristae, and

vacuolization. Metabolically, the activities of key respiratory chain complexes I and IV are severely impaired, leading to inefficient oxidative phosphorylation (OXPHOS) and a significant reduction in adenosine triphosphate (ATP) production. Dynamically, the axonal transport capacity of mitochondria along microtubule and neurofilament tracks is significantly retarded, affecting their proper subcellular localization and energy supply within neurons. In terms of quality control, the mitophagy pathway is impaired, failing to effectively clear dysfunctional mitochondria. In terms of oxidative stress, electron transport chain disorders result in excessive production of ROS such as superoxide, which further exacerbate mtDNA mutations, accumulation of oxidatively damaged proteins, and lipid peroxidation. The combined effects mentioned above cause AD neurons to be in a state of energy starvation continuously, with calcium overload aggravating excitotoxicity, triggering the activation of mitochondrial apoptotic pathways or inflammatory responses mediated by the mtDNA/cGAS-STING pathway, and ultimately accelerating neurodegeneration (Quan et al., 2025).

3.7. Cytomorphological changes

The morphological changes of senescent cells are extremely prominent in the AD brain. Senescent neurons exhibit soma atrophy, reduced dendritic branching, and beading structures formed due to microtubule instability and local energy/calcium imbalance (Richardson et al., 2023). Meanwhile, the density and maturity of dendritic spines decrease, and axonal swelling forms dystrophic neurites (DNs) that surround amyloid plaques, accompanied by perikaryon atrophy, lipofuscin accumulation, and inclusions containing NFTs (Dehkordi et al., 2021). Under electron microscopy, nuclear membrane folding and invagination are exacerbated, rough endoplasmic reticulum and Golgi apparatus are dilated and fragmented, and mitochondria show vacuolar changes (Zheng et al., 2025). Senescent microglia transform from a ramified to an amoeboid morphology, characterized by soma hypertrophy, short and thick processes with reduced numbers, presenting an activated state, and aggregating around plaques to form crown-like clusters (Rachmian et al., 2024; Rim et al., 2024). Senescent astrocytes convert to a reactive phenotype, with significantly enlarged soma, thickened and elongated processes extending towards the lesion area, forming glial scars (Turnquist et al., 2016). These morphological changes are directly related to their enhanced secretion of pro-inflammatory SASP and hyperactive synaptic phagocytosis.

4. The pathogenesis of senescent cells in AD

AD pathogenesis is multifactorial, with pathological senescence of six key central nervous system cell types (microglia, astrocytes, neurons, neural stem cells, oligodendrocytes, endothelial cells) as a critical driver throughout progression. Senescent cells lose normal functions, secrete SASP, induce neuroinflammation, impair neural homeostasis, disrupt the BBB, and hinder brain repair (Fig. 3). Therefore, systematically analyzing the specific changes, interactions of these six key brain cell types during senescence, and their roles in the AD pathological cascade is an indispensable core link for revealing the complex etiology of AD and identifying intervention targets (Table 1).

4.1. Microglia

As the primary innate immune cells in the central nervous system, microglia are responsible for defending against damage in a quiescent state. They become activated when pathogens invade or tissue damage occurs (such as infection, trauma, and ischemia-reperfusion), secreting various cytokines and growth factors (Ransohoff and Brown, 2012). In aged mouse models, microglia exhibit senescent characteristics, including elevated levels of pro-inflammatory factors and telomere shortening (Bachstetter et al., 2011; Flanary et al., 2007; Sierra et al., 2007). Long-term activated microglia in vitro also display senescent

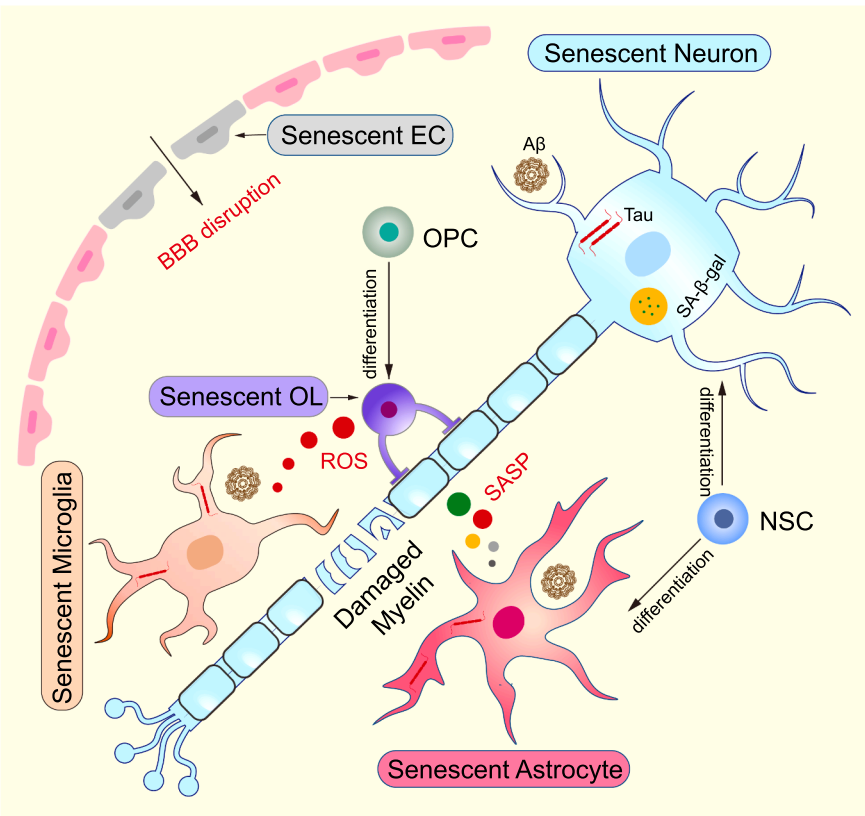


Fig. 3. Schematic representation of cellular senescence heterogeneity and intercellular communication within the Alzheimer's disease (AD) brain microenvironment. The central senescent neuron, characterized by β -galactosidase (β -gal) activity and the presence of pathological markers A β and tau, is surrounded by a variety of senescent cell types, including endothelial cells (ECs) implicated in blood-brain barrier (BBB) dysfunction, oligodendrocytes (OLs) producing reactive oxygen species (ROS), astrocytes, and microglia. Notable features encompass the differentiation behavior of oligodendrocyte progenitor cells (OPCs), compromised myelin integrity, and altered neural stem cell (NSC) function. The coordinated secretion of senescence-associated secretory phenotype (SASP) factors by these senescent cells highlights their collective role in promoting neurodegeneration within the pathological network.

Table 1
Senescence-related characteristics of key brain cell types in Alzheimer's disease (AD).

Rank	Post-Mitotic?	Core Senescence Characteristics	Impact on Brain Function	Key Inducers
Microglia	No	p16 $\uparrow$, p21 $\uparrow$, SASP $\uparrow$, SA- β -gal $\uparrow$, telomere shortening, impaired phagocytosis	Excessive neuroinflammation, reduced A β clearance	Aging, A β aggregates, oxidative stress
Astrocytes	No	p16 $\uparrow$, p21 $\uparrow$, SASP $\uparrow$, SA- β -gal $\uparrow$, enlarged morphology	Disrupted synaptic homeostasis, glial scar formation	A β , ROS, environmental toxins
Neurons	Yes	p16 $\uparrow$, p21 $\uparrow$, γ H2AX $\uparrow$, SASP $\uparrow$, SA- β -gal $\uparrow$, DNA damage, lipofuscin accumulation	Reduced synaptic plasticity, cognitive decline	A β , tau, oxidative stress
Neural Stem Cells	No	p21 $\uparrow$, reduced proliferation, impaired differentiation	Diminished neurogenesis, reduced brain repair capacity	A β oligomers, aging, telomere shortening
Oligodendrocytes	Yes	p16 $\uparrow$, SASP $\uparrow$, SA- β -gal $\uparrow$, reduced myelin synthesis	White matter damage, impaired axonal myelination	Oxidative stress, aging
Endothelial Cells	No	SASP $\uparrow$, tight junction disruption, reduced perfusion	BBB leakage, neuroinflammation, impaired A β efflux	Aging, ROS, vascular dysfunction

markers, such as enhanced SA- β -galactosidase (SA- β -gal) activity, growth arrest, and the formation of SAHF (Yu et al., 2012). This senescent phenotype is similarly present in AD patients: their microglia also have shortened telomeres and show unique dystrophic morphological and functional abnormalities (Conde and Streit, 2006; Flanary et al., 2007; Streit et al., 2009).

Senescent microglia, particularly in their dystrophic phenotype, have increasingly been recognized as key contributors to the pathogenesis of late-onset Alzheimer's disease (LOAD). A growing body of evidence associates their accumulation with the progression of A β and hyperphosphorylated tau pathology, neuroinflammation, and synaptic degeneration. As highlighted by Lau et al. (Lau et al., 2023),

aging-related oxidative stress and AD pathologies promote glial senescence, leading to the secretion of SASP factors by senescent microglia and the propagation of paracrine senescence, thereby accelerating the spread of protein aggregates and advancing Braak staging. Rim et al. (Rim et al., 2024) further identified the TREM2-APOE axis as a critical regulator of microglial senescence, demonstrating that chronic exposure to pathological protein aggregates and myelin debris impairs microglial phagocytic function and metabolic homeostasis. Even so, whether TREM2 upregulation serves as a direct indicator of cellular senescence warrants careful evaluation; its specificity should be validated in conjunction with established senescence biomarkers, and the causal relationship must be further substantiated through additional

experimental evidence. Collectively, these findings position senescent microglia as a central mechanistic link in AD progression, integrating the effects of aging, protein aggregation, and chronic neuroinflammation.

Furthermore, senescence leads to severe impairment of microglial function. Although the levels of TGF- β in the plasma and cerebrospinal fluid of AD patients are increased, the TGF- β 1-SMAD3 signaling pathway in senescent microglia is impaired, resulting in the abnormal loss of their ability to clear A β peptides (Tichauer et al., 2014). This functional defect is associated with the enrichment of senescent microglia in neurodegenerative diseases such as AD and may precede neurodegenerative lesions, suggesting a potential causal relationship between their senescence and disease progression (Conde and Streit, 2006; Streit et al., 2009). The abnormal activation of senescent microglia exerts pro-inflammatory effects by secreting SASP factors and can induce senescence in neighboring cells, thus playing a central role in normal brain aging and neurodegenerative diseases like AD (Yu et al., 2012). It is worth noting that although studies have found that astrocytes in aged mice exhibit a pro-inflammatory phenotype in response to microglial induction, due to the lack of direct detection of senescence markers, whether this pro-inflammatory state is attributed to cellular senescence and SASP remains to be verified (Clarke et al., 2018).

4.2. Astrocytes

Astrocytes are crucial for maintaining brain homeostasis, with functions including providing metabolic support to neurons, regulating water and ion balance as well as neurotransmitter levels, participating in synaptic transmission, and counteracting inflammation related to neurodegeneration, trauma, and infection in the innate immune response (Farina et al., 2007). However, studies have shown that astrocytes undergo senescence, with their numbers increasing in human brain tissue with age, accompanied by elevated levels of senescence markers such as p16, p21, and MMP3 (Bhat et al., 2012; Salminen et al., 2011). Senescent astrocytes exhibit a series of characteristic changes, including enlarged and flattened cell morphology, growth arrest, enhanced SA- β -gal activity, massive production of ROS, reduced mitochondrial activity, and formation of SAHF (Bitto et al., 2010; Evans et al., 2003; Pertusa et al., 2007). Notably, replicative senescence and stress-induced senescence of astrocytes have been found to occur independently of telomere shortening (Bitto et al., 2010). A variety of factors can induce astrocyte senescence: ROS stimulation in vitro can induce senescent phenotypes (Zou et al., 2012), and environmental toxins such as 2,3,7,8-tetrachlorodibenzo-p-dioxin (Nie et al., 2015), ammonia (Görg et al., 2015), and paraquat (Chinta et al., 2018) can effectively trigger their senescence.

Directly related to AD pathology, the proportion of senescent astrocytes in the frontal cortex of AD patients is significantly higher than that in age-matched controls (Bhat et al., 2012). In vitro, primary cultured astrocytes exposed to A β 42 can induce senescence in a concentration-dependent manner (Bhat et al., 2012). Similarly, A β has also been confirmed to trigger astrocyte senescence through the regulation of p38MAPK. A core pathological feature of senescent astrocytes is their secretion of SASP. This includes a significant increase in the secretion of factors such as ICAM-1, GM-CSF, IL-6, and IL-8 through the p38/MAPK signaling pathway after A β exposure (Mombach et al., 2015). Glutathione depletion or oxidative stress can also activate SASP-related pathways NF- κ B and p38MAPK, leading to increased IL-6 secretion (Farina et al., 2007; Hou et al., 2018). Oxidative stress-induced senescence causes significant transcriptomic changes in astrocytes, characterized by downregulation of genes related to nervous system development/differentiation and cell cycle, while upregulation of genes related to inflammation, extracellular remodeling, and anti-apoptosis (Crowe et al., 2016). This senescent state impairs the ability of astrocytes to maintain neuronal function, and their secreted SASP factors have pro-inflammatory effects, which are thought to

participate in the progression of diseases such as AD by accelerating neuropathological processes (Bitto et al., 2010; Evans et al., 2003; Nie et al., 2015). Evidence from in vivo studies shows that progeric BubR1 mutant mice exhibit changes in astrogliosis associated with activated astrocytes (Hartman et al., 2007). Progeroid BubR1 mutant mice represent a pivotal model for investigating premature aging and cellular senescence, harboring mutations in the *BubR1* gene, which plays an essential role in spindle assembly checkpoint-mediated chromosome segregation. The defective BubR1 protein compromises cell cycle fidelity, leading to chromosomal instability and accelerated cellular senescence, as well as postnatal growth retardation, age-related pathologies—including muscle atrophy, vascular deterioration—and a significantly shortened lifespan (Baker et al., 2008; Matsumoto et al., 2007; Wang et al., 2003). By recapitulating senescence-associated multi-organ degeneration, these mice serve as a valuable platform for elucidating the mechanisms of aging and evaluating potential anti-aging therapeutic strategies. This further supports the link between cellular senescence and astrocyte dysfunction in neurodegenerative diseases.

4.3. Neurons

Although neurons, as terminally differentiated post-mitotic cells, do not fully conform to the classical definition of cellular senescence, accumulating evidence indicates that neurons accumulate senescence-like phenotypes during the aging process and play a crucial role in the pathogenesis of AD (Niu et al., 2024). Studies have shown that neurons in aged mice exhibit senescence markers such as increased DNA damage, elevated SA- β -gal activity, activation of p38MAPK, accumulation of IL-6, and lipofuscin (Jurk et al., 2012; Moreno-García et al., 2018; Sedelnikova et al., 2004), with approximately 20–80 % of neurons displaying such phenotypes (Jurk et al., 2012). In AD patients, neurons specifically exhibit characteristics including enhanced p38/MAPK activity, increased expression of TGF- β and IL-6, upregulated p16 expression, and elevated SA- β -gal activity. This senescence-like phenotype was early detected in hippocampal pyramidal cells of rats and the hippocampal dentate gyrus/CA3 region of accelerated aging models (Geng et al., 2010; Ota et al., 2012). Especially, telomere shortening occurs in AD patients, which is significantly different from Parkinson's disease (PD) (Forero et al., 2016a, b). Moreover, the senescence-like phenotype of neurons is exacerbated in a telomere dysfunction model (TERC knockout mice), and this effect is mediated through the p21 pathway, as evidenced by rescue experiments in a *Cdkn1a* knockout background (Jurk et al., 2012).

Such senescent neurons secrete SASP factors to induce senescence in neighboring neurons, microglia, astrocytes, and endothelial cells in a paracrine manner, thereby amplifying the neuroinflammatory cascade and degenerative lesions (Niu et al., 2024). Functional studies have revealed that lipofuscin-positive senescent neurons in the nuclei regulating the sleep-wake cycle show reduced dendritic branching and decreased neurotransmitter synthesis capacity (Panossian et al., 2011). Human disease models (induced by Rett syndrome with MECP2 mutation) further confirm that DNA damage can trigger p53-mediated SASP, providing direct mechanistic evidence for the pathological cascade driven by neuron senescence-like phenotypes in diseases such as AD (Ohashi et al., 2018).

4.4. Neural stem cells

Neural stem cells (NSCs) are present in the embryonic and adult brains. Adult NSCs possess the ability of self-renewal and differentiation, which are crucial for maintaining cognitive functions throughout life and repairing nerve damage (Gage, 2000; Khacho et al., 2019). However, NSCs are prone to senescence under various stimuli. In the pathological context of AD, A β 42 oligomers can induce adult mouse hippocampal NSCs to express senescence markers p16 and SA- β -gal, and significantly inhibit their proliferation and differentiation capabilities

(He et al., 2013). Similarly, NSCs cultured in vitro, after long-term exposure to A β oligomers, also exhibit typical senescent characteristics, including enlarged and flattened cell morphology, increased levels of SA- β -gal and p16, and decreased levels of p-RB (Li et al., 2016). These A β -induced NSC senescence phenomena observed in vitro have been verified in vivo in the dentate gyrus of the APP/PS1 transgenic AD mouse (Ferrón et al., 2004). The characteristics of NSC senescence are not limited to responses to AD-related stimuli, but also include telomere shortening and increased ROS production that generally occur during the aging process (Bose et al., 2010; Ferrón et al., 2009). The consequences of NSC senescence are severe, manifested as impaired self-renewal and differentiation abilities. This functional impairment has been observed in various situations: obesity can lead to the accumulation of senescent glial cells near the lateral ventricles, inhibiting adult neurogenesis and causing psychobehavioral abnormalities (Ogrodnik et al., 2019a). In vivo evidence from progeric BubR1 knockout mice also clearly shows that accelerated cellular senescence impairs the process of adult neurogenesis (Yang et al., 2017). The senescence-like phenotypes exhibited by neural progenitor cells in the white matter lesion areas of patients with multiple sclerosis further demonstrate the universality of neural stem/progenitor cell senescence in neurodegenerative diseases (Nicaise et al., 2019). Particularly, studies have also explored the possibility of intervention. For example, hydroxyurea can induce neural stem cell senescence, while α -glycerophosphoethanolamine treatment can improve senescence-related phenotypes, such as reducing elevated ROS levels, alleviating mitochondrial dysfunction, and restoring partial proliferation capacity (Daniele et al., 2016).

4.5. Oligodendrocytes and oligodendrocyte progenitor cells

Oligodendrocytes are terminally differentiated post-mitotic cells responsible for forming myelin sheaths around axons, but they are highly vulnerable to oxidative stress (Giacci et al., 2018). Neuroimaging and neuropathological evidence indicate significant myelin damage in white matter associated with aging and AD (Brickman, 2013; Erten-Lyons et al., 2013; Gouw et al., 2008). This damage may partially stem from stress-related senescence processes experienced by oligodendrocytes, as oxidative DNA damage and upregulated SA- β -gal activity have been observed in oligodendrocytes of aged individuals, suggesting the occurrence of cellular senescence (Al-Mashhadi et al., 2015; Nasrabady et al., 2018). Unlike oligodendrocytes, oligodendrocyte precursor cells (OPCs) are a type of precursor cells accounting for approximately 3–10 % of glial cells in the adult brain (Dawson et al., 2003), with the ability to mediate remyelination by differentiating into myelinating oligodendrocytes (Reynolds et al., 2002). Although OPCs themselves do not undergo replicative senescence, in vitro studies have shown that they can enter a senescence-like state under certain conditions. A key factor inducing OPC senescence is esophageal cancer-related gene 4 (Ecr4), whose overexpression leads to cell cycle arrest and increased SA- β -gal expression in OPCs (Kujuro et al., 2010). Notably, in the brains of aged mice, both OPCs and NSCs exhibit increased Ecr4 expression levels (Kujuro et al., 2010), suggesting that Ecr4-mediated mechanisms may play a role in aging-related OPC dysfunction and reduced remyelination capacity, particularly in the context of neurodegenerative diseases associated with myelin damage such as AD.

4.6. Endothelial cells

Senescent endothelial cells are a key driver of BBB disruption in AD, and may even precede the emergence of classic A β deposition and NFTs (Nation et al., 2019). Cerebral microvascular endothelial cells form the core structure of the BBB, responsible for regulating substance exchange, maintaining the homeostasis of the intracerebral microenvironment, and participating in neurovascular coupling (Sweeney et al., 2018).

With aging, endothelial cells undergo replicative senescence, characterized by telomere shortening, mitochondrial dysfunction, ROS accumulation, and activation of SASP (Ungvari et al., 2018; Yamazaki et al., 2016). SASP leads to the continuous release of IL-6, TNF- α , IL-1 β , and MMPs, inducing a state of chronic low-grade inflammation, disrupting the expression and distribution of BBB tight junction proteins (claudin-5, occludin, ZO-1), and increasing BBB permeability (Knopp et al., 2023; Rochfort et al., 2015). This leakage not only allows neurotoxic substances such as fibrinogen, coagulation factors, and inflammatory cells in the peripheral circulation to enter the brain parenchyma, exacerbating neuroinflammation, but also impairs the efflux and clearance capacity of A β from the brain to the blood through the low-density lipoprotein receptor-related protein 1 (LRP1), while promoting the influx of A β from the blood to the brain mediated by the receptor for advanced glycation end products. Eventually, this results in abnormal deposition of A β in the brain (Montagne et al., 2017). In addition, oxidative stress and energy metabolism disorders in senescent endothelial cells impair neurovascular coupling function, that is, the coupling between neuronal activity and increased regional cerebral blood flow (CBF), leading to cerebral hypoperfusion and chronic cerebral hypoperfusion. This further limit the energy supply to neurons and accelerates neurodegeneration (Kiss et al., 2020). It is worth noting that senescence of endothelial cells may also activate pericytes and astrocytes by releasing TGF- β , amplifying damage to the neurovascular unit (Ungvari et al., 2018). Recent studies have also revealed impaired autophagic flux and defective mitochondrial quality control in senescent endothelial cells (Yamazaki et al., 2016), which exacerbate intracellular oxidative damage and energy crisis, and affect neighboring cells by transmitting senescence signals through exosomes.

5. Senotherapeutics: mechanistic basis for senescent cells

As a core strategy for intervening in aging-related pathologies, senotherapeutics mainly includes two major approaches: Senolytics aim to selectively induce apoptosis and clearance of senescent cells by targeting their unique survival vulnerabilities (such as the highly dependent anti-apoptotic BCL-2 family, PI3K/AKT/mTOR signaling axis, HSP90 chaperone function, etc.), thereby removing the burden of senescent cells at the source (Table 2). Senomorphics, on the other hand, focus on inhibiting the harmful SASP of senescent cells. They block the release of pro-inflammatory factors and tissue-destructive proteases by precisely regulating core inflammatory signaling hubs such as NF- κ B, JAK-STAT, and p38MAPK, so as to reduce their toxic impact on the microenvironment, rather than directly eliminating the cells themselves (Table 3). Understanding the molecular mechanisms of these two types of strategies is the basis for developing effective anti-aging interventions (Fig. 4).

5.1. Senolytic therapy

5.1.1. Dasatinib and quercetin

- Dasatinib:** As a multi-target tyrosine kinase inhibitor, Dasatinib specifically disrupts the survival-dependent pathways of senescent cells by inhibiting the PI3K/AKT signaling axis and BCL-XL in the BCL-2 anti-apoptotic protein family, thereby inducing their apoptotic clearance. In leukemia models, its role is further manifested in inhibiting LYN, HCK (members of the SRC family kinases) and the receptor tyrosine kinase c-KIT, blocking the proliferation and self-renewal capacity of leukemia stem cells. This kinase-targeted mechanism of cell cycle arrest is also applicable to the clearance of senescent cells (Han et al., 2010). Phosphoproteomic studies have revealed that Dasatinib interferes with the DNA damage repair system by dual inhibition of the SRC/EGFR signaling network and the downstream RAS/MAPK pathway, exacerbates metabolic stress in senescent cells (Li et al., 2010). In the combined treatment of solid

Table 2
A comprehensive list of senolytics targeting cellular senescence.

Rank	Agent	Therapeutic target
1	Dasatinib + Quercetin	Bcr-Abl and Src tyrosine kinase inhibitor PI3K/Akt and mTOR inhibitor
2	Fisetin	PI3K/Akt inhibitor
3	Geldanamycin	HSP90 inhibitors
4	Tanespimycin/ 17-AAG	
5	Alvespimycin/ 17-DMAG	
6	Ganetespib	
7	Navitoclax/ ABT-263	BCL-2 family inhibitors
8	ABT-737	
9	A-1331852 and A-1155463	
10	Panobinostat	
11	Curcumin analog EF24	
12	FOXO4-DRI	p53 pathway modulators
13	UBX0101	
14	RG7112	
15	P5091	
16	P22077	
17	Digoxin	Na ⁺ /K ⁺ -ATPase inhibitors
18	Ouabain	
19	Proscillaridin A	
20	Ouabagenin	
21	Bufalin	
22	K-Strophanthin	
23	Strophanthidin	
24	SSK1	SA-β-galactosidase-targeted prodrugs
25	JHB75B	
26	Nav-Gal	
27	5FURGal	
28	MitoTam	MitoTam
29	CAR-T	Chimeric antigen receptor T-cell therapy
30	Quercetin surface functionalized Fe3O4 nanoparticle	Nanoparticle Based Delivery
31	Antibody-functionalized mesoporous silica nanoparticle	
32	Bone affinity peptide (DSS) modified liposomes loaded with quercetin	
33	Nano-encapsulated navitoclax	

Table 3
A comprehensive list of senomorphics targeting cellular senescence.

Rank	Agent	Therapeutic target
1	Rapamycin	mTOR inhibitors
2	Epigallocatechin gallate (EGCG)	
3	Apigenin and kaempferol	NF-κB inhibitors
4	Metformin	
5	Resveratrol	
6	Piperlongumine and its analogues	
7	SR12343	
8	Curcumin and its metabolite	
	Cortisol and Corticosterone	
9	Oleuropein aglycone	
10	Hydroxytyrosol	
11	SB203580	p38MAPK and MK2 kinase inhibitors
12	UR13756	
13	BIRB796	
14	PF-3644022	
15	MK2.III	
16	Ruxotinilub	JAK inhibitors
17	Momelotinib	
18	KU-60019	ATM kinase inhibitors
19	KU-55933	

tumors, dasatinib inhibits angiogenesis-related receptors such as PDGFRβ, and this inhibitory action together with its targeting of other tyrosine kinases such as SRC family kinases, ABL and c-KIT contributes to its antitumoral effects including inhibiting tumor cell

proliferation, migration and invasion as well as modulating the tumor microenvironment when used in combination with standard-of-care treatments for various solid tumors (Montero et al., 2011).

- **quercetin:** In mechanistic studies on intervening in cellular senescence, quercetin exerts a key senolytic effect by targeting the PI3K/Akt-mTOR signaling axis (Zhu et al., 2015): This compound directly inhibits PI3K kinase activity, blocking the phosphorylation of Akt at Ser473/Thr308 sites, thereby interfering with downstream pro-survival pathways. Meanwhile, quercetin binds to the ATP pocket domain of mTOR, inhibiting the activity of the mTORC1 complex, leading to the dephosphorylation of its downstream effectors 4EBP1 and S6K, and thus blocking protein anabolism. This dual pathway inhibition synergistically impairs the anti-apoptotic defense network of senescent cells, particularly targeting the BCL-2/BCL-XL-dependent survival mechanism, thereby selectively inducing apoptosis in senescent cells.
- **D+Q:** The combination of Dasatinib and quercetin (D+Q) intervenes in cellular senescence through a multi-target synergistic mechanism: In neurodegenerative disease models, D+Q selectively clears senescent microglia and astrocytes expressing p16. By inhibiting the PI3K/AKT-BCL-2 anti-apoptotic pathway, it activates caspase-3-dependent apoptosis, significantly reduces tau protein phosphorylation and neurofibrillary tangles in the brain, and improves cognitive function (Bussian et al., 2018). In aging-related metabolic diseases, this combination inhibits the release of SASP factors by blocking the mTOR-NF-κB signaling axis, alleviates hepatic lipid accumulation and obesity-induced neurogenesis impairment in the hippocampus (Ogrodnik et al., 2017, 2019b), and reshapes the intestinal micro-environment. This shows increasing the abundance of *Akkermansia*, inhibiting the infiltration of SA-β-gal positive cells, and down-regulating TLR4/MyD88 mediated IL-1β secretion (Saccon et al., 2021). In the field of tissue fibrosis, D+Q reduces the expression of PAI-1 and MMP9 in senescent alveolar epithelial cells by interfering with the p38MAPK-FOXO1 pathway, thereby delaying the progression of pulmonary fibrosis (Schafer et al., 2017). In bone metabolism research, it restores the mineralization ability of senescent osteoblasts by inhibiting JAK2-STAT3 phosphorylation, leading to an increase in trabecular bone volume (Farr et al., 2017). Especially, the combination of D+Q exerts a critical regulatory effect on tau-related pathology: beyond directly clearing senescent cells, they disrupt the vicious cycle between NFTs formation and cellular senescence, and in turn reduce NFTs density, neuronal loss, and brain atrophy in tau transgenic mouse models (Musi et al., 2018). The core of this synergistic mechanism lies in Dasatinib targeting tyrosine kinase-dependent survival pathways, while quercetin strengthens oxidative stress defense and inhibits SASP epigenetic regulators, collectively reshaping the overall homeostasis of the aging microenvironment.

To date, clinical research on D+Q for AD and AD-related risk states includes completed phase 1/pilot trials and an ongoing phase II trial (Table 4). The phase 1 trial (NCT04063124) and its biomarker analysis showed 12-week D+Q was well-tolerated with detectable dasatinib in CSF, altered fluid biomarkers (e.g., increased CSF IL-6/GFAP, plasma fractalkine/MMP-7; downregulated inflammatory CTRA genes in PBMCs; reduced plasma phosphatidylcholine/acylcarnitine), while CSF Aβ/tau and cognitive/neuroimaging outcomes stayed stable (Garbarino et al., 2025; Gonzales et al., 2023), they failed to show significant cognitive benefits. The pilot trial (NCT05422885) confirmed D+Q's feasibility/safety in at-risk older adults, with preliminary cognitive benefits (2.0-point MoCA increase in low-baseline participants, inversely correlated with lower TNF-α). These findings are limited by small samples (n = 5–12), 12-week durations, open-label design, and restricted generalizability (predominantly white, highly educated population (Lewis, 2022; Millar et al., 2025)). An ongoing phase II trial

Anti-senescence strategies

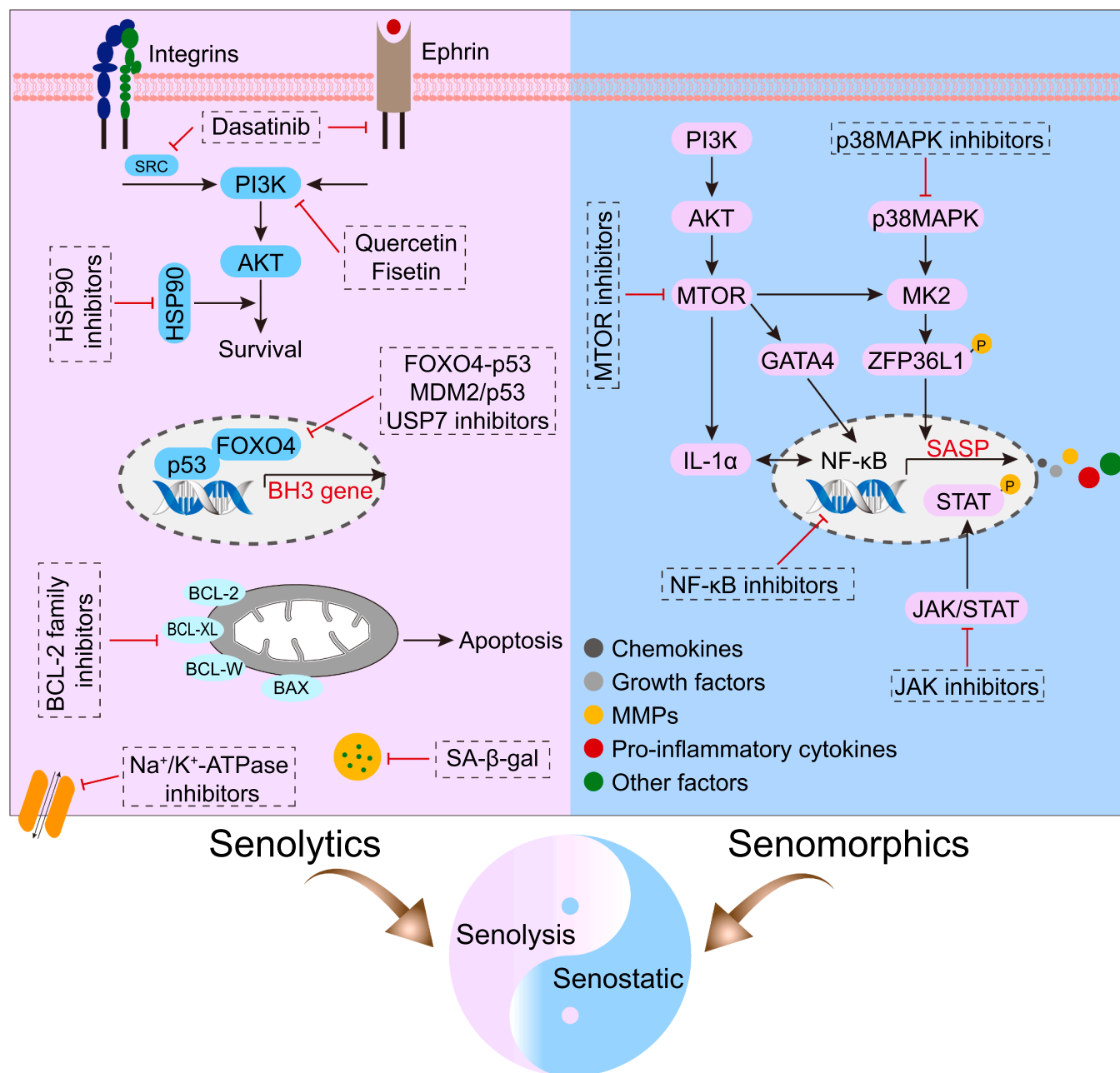


Fig. 4. Schematic overview of pharmacological strategies targeting cellular senescence in Alzheimer's disease (AD), categorizing interventions into senolytics (promoting senescent cell elimination, senolysis) and senomorphics (modulating senescence-associated phenotypes, senostatic). Senolytic approaches (left) act through multiple pathways: integrin-src-PI3K-AKT for survival modulation, HSP90 inhibition, disruption of FOXO4-p53 interactions, inhibition of BCL-2/BCL-XL/BCL-W within apoptosis pathways to induce apoptosis, and Na⁺/K⁺-ATPase inhibition to reduce SA-β-gal, with relevant agents and molecular targets depicted. Senomorphics strategies (right) target suppression of key pro-inflammatory signaling cascades for SASP production, including modulation of PI3K-AKT-mTOR, p38MAPK-MK2, NF-κB, and JAK-STAT pathways, collectively reducing SASP components (pro-inflammatory cytokines, MMPs, chemokines, growth factors) to exert senostatic effects. Arrows denote molecular targets and pathway relationships, illustrating therapeutic avenues to counter cellular senescence, with potential relevance for AD and other age-related pathologies.

(NCT04685590, 24 months) aims to evaluate cognitive decline rate (ADAS-Cog), brain atrophy, and SASP levels, to provide more robust evidence of D+Q's long-term safety and potential disease-modifying effects (Washington University School of Medicine, 2021). In summary, existing evidence indicates that D+Q holds promise in improving cognitive function, activities of daily living, and physical function in AD patients and high-risk populations by clearing senescent cells, reducing neuroinflammation, and improving pathological biomarkers. Its safety

and long-term benefits are being verified in more in-depth clinical stages.

5.1.2. PI3K/Akt inhibitor

Fisetin, a natural flavonoid that targets the PI3K/Akt signaling pathway, blocks downstream mTOR and NF-κB signal transduction by directly inhibiting the excessive activation of the PI3K/Akt pathway, thereby reducing the expression of anti-apoptotic proteins such as BCL-2

Table 4
Trials of dasatinib and quercetin (D+Q) as senolytics in Alzheimer's disease (AD).

Rank	Title	Status	Intervention	Primary Outcomes	Limitations
1	Senolytic therapy in mild Alzheimer's disease: a phase 1 feasibility trial (NCT04063124)	Completed	D+Q (12 weeks)	- Well-tolerated with detectable D in 5 participants' CSF, Q undetectable in CSF; - Significant CSF IL-6 and GFAP increase; no significant cognitive/neuroimaging changes	- Small sample size (n = 5) and open-label design; - Short 12-week duration, insufficient to assess long-term safety or disease-modifying effects
2	Evaluation of exploratory fluid biomarkers from a phase 1 senolytic trial in mild Alzheimer's disease (NCT04063124)	Completed	D+Q (12 weeks)	- Significantly increased plasma fractalkine, plasma MMP-7, and CSF IL-6; downregulated 7 inflammatory CTRA genes (FOSB, PTGS2, IL8, FOS, IL1 β , JUNB, JUN) in peripheral blood mononuclear cells (PBMCs) post-treatment; - Stable CSF A β /tau levels, unchanged urinary metabolites, and reduced plasma phosphatidylcholine (PC), lysophosphatidylethanolamine (LPE), and acylcarnitine post-treatment.	- Small sample size (n = 5) - Short 12-week duration
3	A pilot study of senolytics to improve cognition and mobility in older adults at risk for Alzheimer's disease (NCT05422885)	Completed	D+Q (12 weeks)	- Feasible and safe: ~10 % telephone-screened participants completed the trial (n = 12), no serious adverse events related to D+Q; - Preliminary cognitive benefit: MoCA scores increased significantly by 2.0 points in participants with lowest baseline MoCA; TNF- α reduction correlated inversely with MoCA increases	- Small sample size (n = 12); - Limited generalizability: study population was predominantly white and highly educated, not representative of broader older adults with MCI and slow gait.
4	Phase II Clinical Trial to Evaluate the Safety and Feasibility of Senolytic Therapy in Alzheimer's Disease (NCT04685590)	Recruiting	D+Q (24 months)	- Primary: Rate of cognitive decline (ADAS-Cog); - Secondary: Brain atrophy, SASP levels	Ongoing, no results yet

and BCL-XL. Meanwhile, it activates the p53-dependent apoptotic pathway, promotes the mitochondrial translocation of pro-apoptotic proteins BAX/BAK, and synergistically breaks down the survival barriers of senescent cells. This mechanism has been verified in various models: in rapidly aging SAMP8 mice, fisetin significantly improves cognitive decline and physiological function degradation, and its effect is related to the clearance of senescent cells in the brain and the restoration of neuronal plasticity (Currais et al., 2018). In the cardiovascular system, it can alleviate doxorubicin-induced cardiotoxicity, and reduce fibrosis and mitochondrial dysfunction by clearing senescent cardiomyocytes (Lérida-Viso et al., 2022). At the vascular endothelial level, fisetin selectively clears senescent subsets of EA.hy926 and HUVEC cells, but the differences in their sensitivity suggest the existence of tissue-specific mechanisms (Abdelgawad et al., 2022). It is worth noting that the broad-spectrum anti-aging effect of fisetin also stems from its in-depth regulation of SASP — by inhibiting the Akt/I κ B kinase complex, it reduces the release of pro-inflammatory factors such as IL-6 and TNF- α , thereby breaking the inflammatory vicious cycle in the senescent microenvironment (Yousefzadeh et al., 2018). Fisetin has both multi-target kinase inhibition and natural antioxidant properties. In cancer models, it can also inhibit tumor progression by blocking EMT transformation and angiogenesis (Syed et al., 2016). Ultimately, by coordinating senescence clearance, inflammation alleviation, and metabolic homeostasis reconstruction, it extends healthy lifespan while reducing the risk of multi-organ degenerative diseases.

5.1.3. HSP90 inhibitors

Geldanamycin and its derivatives, including Tanespmycin (17-AAG), Alvospimycin (17-DMAG), and Ganetespib, act as HSP90 inhibitors. Their core mechanism in intervening in cellular senescence lies in selectively disrupting the protein homeostasis of senescent cells and triggering apoptosis by inhibiting the chaperone function of HSP90. Specifically, these drugs tightly bind to the N-terminal ATP-binding pocket of HSP90, blocking its ATPase activity, which prevents HSP90 from maintaining the correct folding and stability of client proteins (Roe et al., 1999). This process is particularly lethal to senescent cells because they are highly dependent on HSP90 to stabilize a series of pro-survival signaling proteins such as PI3K, Akt, and mTOR. When HSP90 function is inhibited, these key proteins undergo rapid ubiquitination and degradation, leading to the collapse of the PI3K/Akt survival signaling

pathway. On the other hand, HSP90 inhibition induces an increase in mitochondrial membrane permeability, releasing cytochrome c and activating the caspase cascade reaction, ultimately resulting in the selective clearance of senescent cells with minimal impact on normal cells (Fuhrmann-Stroissnigg et al., 2017). This mechanism has been verified in various senescence models as a novel senolytic strategy, which alleviates tissue dysfunction and delays aging-related pathological processes by clearing senescent cells.

5.1.4. BCL-2 family inhibitors

- **Navitoclax:** as konw as ABT-263, an oral BH3 mimetic that targets the anti-apoptotic proteins BCL-2, BCL-xL, and BCL-w, exerts its core mechanism in intervening in cellular senescence by selectively dismantling the unique survival barriers of senescent cells. By competitively binding to the BH3 domain of BCL-xL, the drug prompts the release of pro-apoptotic proteins BAX/BAK from the inhibition of BCL-xL, which then form pores in the outer mitochondrial membrane, trigger the release of cytochrome c, activate the caspase cascade, and ultimately induce apoptosis in senescent cells. This targeted clearance mechanism has been confirmed in multiple organ systems: in the hematopoietic system, ABT-263 eliminates senescent hematopoietic stem cells and restores youthful regenerative capacity (Chang et al., 2016). In lung tissue, it selectively clears senescent type II alveolar cells and reverses radiation-induced pulmonary fibrosis (Pan et al., 2017). In the nervous system, the clearance of senescent glial cells can inhibit pathological deposition of tau protein and improve cognitive function (Bussian et al., 2018). In the cardiovascular field, it targets and removes senescent cardiomyocytes, alleviating cardiac dysfunction (Anderson et al., 2019). In atherosclerosis, the clearance of senescent foam cells in blood vessels can inhibit plaque progression (Childs et al., 2016). Notably, the high sensitivity of senescent cells to ABT-263 stems from their reliance on BCL-xL to maintain an anti-apoptotic state, while normal cells have higher tolerance due to the expression of redundant anti-apoptotic proteins such as MCL-1 (Mérino et al., 2012). However, this selectivity has limitations. For example, toxicity to platelets arises because platelets depend on BCL-xL for survival, leading to transient thrombocytopenia (Schoenwaelder et al., 2011). At the clinical translation level, trials of ABT-263 in combination with

dasatinib/quercetin for the treatment of pulmonary fibrosis have further verified the feasibility of the senolytic strategy (Nambiar et al., 2023). Meanwhile, its role in the post-chemotherapy micro-environment suggests that clearing senescent cells can reduce the risk of cancer recurrence (Demaria et al., 2017).

- **ABT-737:** ABT-737, an oral BH3 mimetic targeting BCL-2, BCL-XL, and BCL-w, shares a similar core mechanism with ABT-263 in intervening in cellular senescence. In the context of senescence intervention, ABT-737 has been proven to clear senescent hematopoietic stem cells and restore their regenerative capacity. In diabetic models, it can eliminate senescent pancreatic islet β cells, thereby delaying the progression of type I diabetes (Thompson et al., 2019). Notably, its effects exhibit significant tissue selectivity: in solid tumors, it induces apoptosis of tumor cells through a similar mechanism (Oltersdorf et al., 2005); while in chronic lymphocytic leukemia, its pro-apoptotic effect depends on the survival dependence of tumor cells on BCL-2, which is achieved by tumor cells sequestering the pro-apoptotic protein BIM (Del Gaizo Moore et al., 2007). Although ABT-737 was optimized into Navitoclax (ABT-263) due to its low oral bioavailability (Tse et al., 2008), both share the core mechanism: by targeting the overexpressed BCL-XL/BCL-w network in senescent cells, they specifically disrupt their anti-apoptotic defense, with limited impact on cells with high MCL-1 expression (van Delft et al., 2006). This characteristic lay the molecular foundation for its role as a senolytic drug, ultimately achieving the goal of selectively clearing senescent cells to improve tissue dysfunction.
- **A-1331852 and A-1155463:** The BCL-XL-specific inhibitors A-1331852 and A-1155463 exert anti-aging effects by targeting the survival-dependent mechanism of senescent cells: These drugs bind with high affinity to the BH3 domain of BCL-XL protein, precisely mimicking the functions of endogenous pro-apoptotic proteins (BIM and PUMA), and competitively block the binding of BCL-XL to pro-apoptotic effector proteins BAX/BAK, thereby relieving the inhibition of BCL-XL on the mitochondrial apoptotic pathway. This mechanism is particularly effective in senescent cells because senescent cells significantly upregulate BCL-XL expression to resist apoptotic stress for survival maintenance. The inhibitors selectively disrupt this "senescence-associated anti-apoptotic barrier", prompting BAX/BAK to oligomerize and form pores in the outer mitochondrial membrane, triggering the release of cytochrome c and activating the caspase cascade, ultimately inducing apoptosis in senescent cells (Zhu et al., 2017). In liver fibrosis models, this mechanism has been confirmed to simultaneously clear senescent cholangiocytes and activated fibroblasts—both of which rely on BCL-XL for survival. By eliminating pro-fibrotic signals (TGF- β and IL-11) derived from these senescent cells and reducing extracellular matrix deposition, it significantly improves liver fibrosis lesions in Mdr2^{-/-} mice (Moncsek et al., 2018).
A1331852 and A1155463 have exhibited fewer adverse effects on neutrophil levels compared to navitoclax, thus earning them characterization as safer compounds (Levenson et al., 2015; Zhu et al., 2017); notably, as BCL-XL-specific inhibitors, they retain senolytic efficacy by targeting BCL-XL-dependent senescent cells, while their minimized hematological impact further supports their potential for translational research in age-related pathologies where mitigating treatment-related toxicities is essential.
- **Panobinostat:** Panobinostat as a pan-histone deacetylase (HDAC) inhibitor, exerts its core mechanism in intervening in cellular senescence by selectively dismantling the survival barriers of senescent cells through epigenetic reprogramming. It binds with high affinity to the catalytic domains of class I and II HDACs (including HDAC1, 2, 3, and 6), blocking their deacetylase activity, which leads to hyperacetylation of intracellular histone substrates (such as H3 and H4) and non-histone substrates (such as p53 and HSP90). In senescent cells, this epigenetic perturbation triggers a cascade of

effects: histone hyperacetylation opens the chromatin structure, activating the transcription of pro-apoptotic genes (such as NOXA and BIM). Meanwhile, acetylated p53 enhances its stability and transcriptional activity, thereby upregulating the expression of the pro-apoptotic protein PUMA, while the key survival factors BCL-2 and BCL-XL are inhibited. On the other hand, the inactivation of HDAC6 leads to the accumulation of misfolded proteins, exacerbating endoplasmic reticulum stress and activating the unfolded protein response (UPR), which synergistically promotes the mitochondrial apoptotic pathway. Particularly, senescent cells are highly sensitive to HDAC inhibition because they rely on HDACs to maintain DNA damage repair mechanisms and suppress endogenous apoptotic signals, whereas normal cells have stronger tolerance due to redundant anti-apoptotic mechanisms (Samaraweera et al., 2017). This selectivity enables panobinostat to efficiently clear therapy-induced senescent cells in non-small cell lung cancer and head and neck squamous cell carcinoma, blocking the formation of their tumor-promoting microenvironment. In addition, its role in multiple myeloma also stems from a similar mechanism — interfering with DNA repair and protein homeostasis in cancer cells through HDAC inhibition (Cheng et al., 2015).

- **Curcumin analog EF24:** As a curcumin derivative, the core mechanism of EF24 in intervening in cellular senescence lies in the selective induction of apoptosis in senescent cells through proteasome-mediated degradation of the Bcl-2 anti-apoptotic protein family (Li et al., 2019). Specifically, EF24 significantly reduces the levels of these key anti-apoptotic proteins by promoting the proteasome-dependent degradation of Bcl-xl and Mcl-1 proteins in senescent cells, and this process is independent of transcriptional regulation. Since senescent cells are highly dependent on Bcl-xl and Mcl-1 to maintain a survival barrier against apoptosis, their degradation leads to the irreversible activation of the mitochondrial apoptotic pathway — the pro-apoptotic proteins BAX/BAK are released and oligomerized, triggering the caspase cascade reaction, and ultimately achieving the precise clearance of senescent cells. Notably, EF24 has a broad-spectrum effect: it exhibits significant killing efficacy on senescent cells from various tissue sources (human lung fibroblasts WI-38/IMR-90, vascular endothelial cells HUVEC, renal epithelial cells HREC, preadipocytes) and those induced by different stresses (replicative senescence, radiation-induced senescence, Ras oncogene-induced senescence), but has little impact on normal cells. Meanwhile, this mechanism is independent of the ROS pathway and can synergistically enhance the senolytic effect with the BCL-2/BCL-xL inhibitor ABT-263.

5.1.5. p53 pathway modulators

- **FOXO4-DRI:** The core mechanism of FOXO4-DRI as a targeting agent for senescent cells lies in its specific blocking of the interaction between FOXO4 and p53, thereby selectively inducing apoptosis in senescent cells. In senescent cells, the FOXO4 protein forms a complex by binding to p53, retaining p53 in the cell nucleus and inhibiting its pro-apoptotic function. FOXO4-DRI mimics the conserved domain of FOXO4 (especially amino acids 49–54), competitively binds to the transcriptional activation domain of p53, thus disrupting the FOXO4-p53 complex (Baar et al., 2017). This promotes the translocation of p53 from the nucleus to mitochondria, thereby activating the BAX/BAK-dependent mitochondrial apoptotic pathway, releasing cytochrome c, and triggering the caspase cascade reaction. In senescent Leydig cells, this mechanism effectively clears dysfunctional senescent cells, restores testosterone secretion capacity, and improves age-related sex hormone deficiency (Zhang et al., 2020). In models of chemotherapy-induced damage and natural aging, FOXO4-DRI significantly reduces fibrosis and inflammatory responses by clearing the burden of senescent cells in tissues, and reconstructs tissue homeostasis (Baar et al., 2017).

- **UBX0101 and RG7112:** UBX0101 and RG7112/RO5045337, as inhibitors targeting the MDM2-p53 interaction, exert their core mechanism in intervening in cellular senescence by stabilizing p53 protein levels and selectively eliminating the apoptotic resistance of senescent cells. In senescent cells, MDM2 is continuously highly expressed and binds to p53, mediating the ubiquitination and degradation of p53, thereby blocking the p53-dependent apoptotic pathway. UBX0101 specifically inhibits the binding of MDM2 to p53, blocks the ubiquitination process, stabilizes the p53 protein that is elevated due to the accumulation of DNA damage, and then activates the downstream pro-apoptotic proteins PUMA and BAX, triggering the mitochondrial apoptotic pathway. It effectively clears senescent chondrocytes in a post-traumatic osteoarthritis model, reduces inflammatory responses and promotes cartilage tissue regeneration (Jeon et al., 2017). As a small-molecule antagonist of MDM2, RG7112/RO5045337 restores p53 activity by binding to the p53-binding pocket of MDM2. It clears senescent nucleus pulposus cells in a human intervertebral disc degeneration model, significantly alleviating intervertebral disc degeneration and related pain (Cherif et al., 2020).
- **P5091 and P22077:** The USP7 inhibitors P5091 and P22077 selectively restore the pro-apoptotic function of p53 to clear senescent cells by targeting the unique protein homeostasis imbalance in senescent cells. In senescent cells, the deubiquitinase USP7 maintains the survival barrier through a dual mechanism: on one hand, it stabilizes the E3 ubiquitin ligase MDM2 to promote p53 degradation (Chauhan et al., 2012; He et al., 2020); on the other hand, it deubiquitinates and stabilizes DNMT1, thereby achieving epigenetic inhibition of p53 target genes. P5091/P22077 block the deubiquitinating activity of USP7 by binding to its catalytic domain with high affinity. This not only weakens the p53 degradation pathway by inducing ubiquitination and degradation of MDM2, but also relieves the transcriptional inhibition of p53 by triggering ubiquitination and degradation of DNMT1, thus synergistically enhancing the stability and transcriptional activity of p53 protein. The activated p53 then upregulates the expression of pro-apoptotic proteins PUMA and BAX, and synergistically inhibits the anti-apoptotic members of the BCL-2 family, ultimately triggering the mitochondrial apoptotic pathway. This mechanism is highly selective for senescent cells, as they rely on USP7 to maintain p53 inhibition to resist apoptotic pressure induced by accumulated DNA damage. In contrast, normal cells have significantly higher tolerance due to the steady-state regulation of the p53 pathway and low levels of DNA damage.

5.1.6. Na^+/K^+ -ATPase inhibitors

Cardiac glycosides (including Digoxin, Ouabain, Proscillaridin A, Ouabagenin, Bufalin, K-Strophanthin, and Strophanthidin) achieve selective clearance by targeting the ionic homeostasis vulnerability of senescent cells. Their core mechanism lies in inhibiting the endoplasmic reticulum stress-mitochondrial apoptosis cascade induced by Na^+/K^+ -ATPase (Guerrero et al., 2019; Triana-Martínez et al., 2019): these drugs bind to the α -subunit of Na^+/K^+ -ATPase with high affinity, block its ion transport function, lead to an increase in intracellular Na^+ concentration, activate the reverse sodium-calcium exchanger (NCX), and drive massive Ca^{2+} influx. In senescent cells, the accumulated Ca^{2+} significantly exacerbates endoplasmic reticulum stress, activates the IRE1 α -XBP1 pathway, and selectively upregulates the expression of the pro-apoptotic protein NOXA. Highly expressed NOXA antagonizes the anti-apoptotic protein MCL-1 (rather than BCL-2 or BCL-xL), releases BAX/BAK to form pores in the outer mitochondrial membrane, and directly activates caspase-3, synergistically triggering the apoptotic program. This mechanism is broad-spectrum, effective in senescence models induced by radiation, replicative stress, and oncogenes. Its selectivity stems from the unique ionic homeostasis imbalance in senescent cells—their membrane potential depolarization makes Na^+/K^+ -ATPase inhibition more likely to cause Ca^{2+} overload, while

normal cells have higher tolerance due to strong ion buffering capacity and low basal levels of NOXA.

5.1.7. SA- β -galactosidase-targeted prodrugs

SA- β -gal-targeted prodrugs (including SSK1, JHB75B, Nav-Gal, and 5FURGal) selectively eliminate senescent cells through an enzyme-triggered precise delivery mechanism. These prodrugs, under the catalysis of β -gal highly expressed in senescent cells, hydrolyze their modified galactose groups to in situ release active toxins: the cytotoxin released by SSK1 directly damages DNA structure (Cai et al., 2020); the scilliroside derivative activated by JHB75B induces DNA alkylation damage (Guerrero et al., 2020); Nav-Gal specifically releases the BCL-2/BCL-xL inhibitor Navitoclax, which triggers BAX/BAK-dependent mitochondrial apoptosis by antagonizing anti-apoptotic proteins (González-Gualda et al., 2020); and the 5-fluorouracil metabolite (FdUMP) generated by 5FURGal inhibits thymidylate synthase, blocking DNA replication (Doan et al., 2020). The core advantage of this strategy lies in utilizing the high activity of lysosomal β -gal unique to senescent cells, enabling the prodrugs to be activated only within senescent cells, thereby significantly reducing off-target toxicity (e.g., the toxicity of Nav-Gal to platelets is 10 times lower than that of Navitoclax). In aged mouse models, this mechanism effectively clears senescent cells in multiple tissues (muscle, brain, joints), reduces the release of inflammatory factors such as IL-6 and TNF- α , and simultaneously improves physical function, cognitive ability, and joint regeneration, providing a new paradigm for the safe intervention of aging-related dysfunctions (Cai et al., 2020).

5.1.8. MitoTam

MitoTam achieves selective elimination by precisely disrupting the mitochondrial energy metabolism homeostasis of senescent cells (Hubackova et al., 2019): Through its lipophilic triphenylphosphonium (TPP⁺) structure, the drug is targeted and enriched in the inner mitochondrial membrane of senescent cells, directly inhibiting the activity of electron transport chain complex I. This leads to the collapse of mitochondrial membrane potential, explosive accumulation of ROS, and dysfunction of OXPHOS. This process triggers the abnormal activation of adenine nucleotide translocase 2 (ANT2), promoting the continuous opening of the mitochondrial permeability transition pore (mPTP). It synergistically exacerbates ATP depletion and the release of cytochrome c, ultimately clearing senescent cells through the caspase-9/3-dependent apoptotic pathway. The selectivity of this mechanism stems from the unique metabolic vulnerability of senescent cells—they rely on high OXPHOS activity for survival, and the threshold for ANT2-mediated mPTP opening is lower. As a result, MitoTam efficiently reduces tissue inflammation and delays functional decline in aged models.

5.1.9. Chimeric antigen receptor T-cell therapy

Chimeric Antigen Receptor T-cell (CAR-T) therapy targeting the urokinase-type plasminogen activator receptor (uPAR) achieves precise immune clearance of senescent cells by genetically engineering T cells to express single-chain antibodies (scFv) that specifically target the senescent cell surface marker uPAR (Amor et al., 2020). In senescence-related pathological models, these CAR-T cells recognize senescent cells with high uPAR expression, such as lung fibroblasts and hepatic stellate cells, and activate T-cell cytotoxic pathways involving the release of perforin/granzyme B and Fas/FasL apoptotic signaling, thereby selectively eliminating senescent cells without damaging normal tissues. Meanwhile, this therapy inhibits the activation of MMP-2 and MMP-9 as well as the degradation of extracellular matrix by blocking the uPAR-mediated plasminogen activation system, alleviating the process of tissue fibrosis. It also downregulates the secretion of IL-6 and PAI-1 driven by TGF- β -SMAD3 signaling, ultimately significantly reversing senescence-related pathological phenotypes in models of pulmonary fibrosis and non-alcoholic steatohepatitis.

5.1.10. Nanoparticle based delivery

Traditional anti-aging drugs face multiple delivery barriers in clinical translation: small-molecule drugs lack tissue targeting, leading to systemic exposure and dose-limiting toxicity. Nucleic acid drugs are easily degraded by enzymes and difficult to penetrate biological barriers, while some polyphenolic compounds have problems such as poor water solubility and low bioavailability. More critically, senescent cells show spatial heterogeneous distribution in tissues, such as specific brain regions, articular cartilage, or vascular endothelium. Systemic administration cannot achieve precise positioning, and traditional formulations are difficult to respond to microenvironmental signals for controlled release. These limitations seriously restrict drug efficacy and increase the risk of off-target effects, highlighting the urgent need to develop new delivery systems to break through biological barriers. Nanocarriers significantly enhance the tissue selectivity and bioavailability of anti-aging drugs through targeted delivery and controlled release mechanisms. In quercetin-functionalized Fe₃O₄ nanoparticles, 3-deazadenosine (3DA) is delivered to senescent cells. It reduces methylation inhibition by inhibiting S-Adenosyl homocysteine hydrolase (SAHH), activates the AMPK/mTOR pathway to improve mitophagy, and quercetin's antioxidant properties synergistically alleviate oxidative damage (Guerrero et al., 2022). Antibody-functionalized mesoporous silica nanoparticles target vascular endothelial cells to deliver the Sirt7 gene, reversing vascular endothelial dysfunction in progeria models and extending lifespan by deacetylating histone H3K18ac and inhibiting the NF- κ B pathway (Sun et al., 2020). Chiral Cu_xCo_yS nanoparticles release Navitoclax under the dual regulation of near-infrared light and magnetic fields, triggering mitochondrial apoptosis in senescent cells through BCL-2/BCL-xL inhibition, and their chiral structure enhances penetration into lysosomes of senescent cells (Li et al., 2020). Liposomes modified with bone-affinity peptides (DSS) preferentially accumulate in bone tissue, sustainably releasing quercetin and resveratrol. They inhibit senescence of lens epithelial cells by activating the Nrf2/ARE pathway and delay the progression of cataracts (Chen et al., 2022). These strategies collectively break through the off-target toxicity limitations of traditional drugs through spatiotemporally precise drug delivery, tissue-specific targeting, and synergistic mechanisms, providing a new paradigm for multi-organ aging intervention.

5.2. Senomorphic therapy

5.2.1. mTOR inhibitors

- **Rapamycin:** As a core hub for regulating aging (Papadopoli et al., 2019), the mTOR signaling pathway drives pro-inflammatory SASP by promoting IL-1 α translation, accelerating organismal aging, and forming a tumor-promoting microenvironment (Labege et al., 2015). Rapamycin can directly block this pathway, inhibit SASP through a mechanism independent of Nrf2, restore autophagic function, and clear senescent cells (Wang et al., 2017; Weichhart, 2018). In mammalian models, rapamycin can significantly extend lifespan, increasing the maximum lifespan of cancer-prone mice by nearly 30 %, while delaying the progression of age-related diseases (Anisimov et al., 2010). This effect stems from the central regulatory role of mTOR in cellular energy metabolism, stress response, and autophagy (Bjedov and Rallis, 2020). However, its clinical application faces adverse reactions such as increased infection risk due to immunosuppression and metabolic disorders (Anisimov et al., 2010; Mao et al., 2022), as well as drug resistance caused by long-term administration (Xu et al., 2020). To overcome these limitations, new-generation highly selective inhibitors such as GDC-0077 (Inavolisib) significantly improve targeting specificity and reduce off-target toxicity by precisely degrading mutant PI3K α proteins (Hanan et al., 2022). Based on existing evidence, targeting the mTOR pathway demonstrates triple value in intervening in aging and related neurodegenerative diseases: mechanistically, it directly

regulates two core aging processes, SASP and autophagy (Labege et al., 2015; Weichhart, 2018); in terms of efficacy, it has been verified in mammalian models to extend lifespan and delay diseases (Anisimov et al., 2010); in translation, the new-generation inhibitors balance efficacy and safety (Hanan et al., 2022). Future efforts need to further explore tissue-specific delivery strategies to address systemic adverse reactions.

- **Epigallocatechin gallate (EGCG):** Epigallocatechin gallate (EGCG), the main active component of green tea polyphenols, exerts dual anti-aging effects by targeting and regulating aging-related signaling pathways. In terms of inhibiting cellular premature senescence, EGCG can effectively block the excessive activation of the PI3K/Akt/mTOR cascade (Kumar et al., 2019): it reduces the phosphorylation level of PI3K, inhibits the activity of downstream Akt and mTOR proteins, thereby reversing the stress-induced premature senescence phenotype of preadipocytes and maintaining the normal proliferation and differentiation functions of cells. This intervention on the growth signaling pathway directly delays the process of cell cycle arrest, and the activity of senescence-associated β -gal is significantly reduced. In terms of clearing senescent cells, EGCG selectively induces apoptosis of senescent cells by regulating the balance of Bcl-2 family proteins. It significantly upregulates the expression of the pro-apoptotic factor Bax and simultaneously inhibits the function of the anti-apoptotic protein Bcl-2, leading to the collapse of mitochondrial membrane potential and the activation of the caspase cascade reaction, ultimately achieving the programmed clearance of senescent cells (Han et al., 2012). Notably, in the human dermal fibroblast model, EGCG can also activate SIRT1 deacetylase, inhibit the acetylation modification of the K382 site of the p53 protein, block the replicative senescence process mediated by the p53-p21/Rb pathway, and reduce oxidative stress damage caused by ROS accumulation (Han et al., 2012). This multi-target synergistic mechanism enables EGCG not only to prevent premature cellular senescence but also to actively clear the established senescent cell pool, providing dual protection for delaying tissue degeneration. Particularly in the field of skin aging, EGCG protects collagen synthesis in the dermis and inhibits the expression of MMPs through the above pathways, laying a molecular foundation for the development of natural preparations against skin photoaging.

5.2.2. NF- κ B inhibitors

- **Apigenin and kaempferol:** Apigenin, a representative flavonoid, directly inhibits the transcriptional activation and release of key pro-inflammatory factors such as IL-6 and IL-8 in the SASP by specifically blocking the nuclear translocation of the NF- κ B p65 subunit, thereby disrupting its paracrine tumor-promoting effect on breast cancer cells (Perrott et al., 2017). This mechanism effectively blocks the malignant signal crosstalk between senescent cells and cancer cells in the pathological microenvironment, significantly reducing the risk of tumor invasion. Kaempferol, a flavonoid abundant in the seed extract of *Nephelium lappaceum* L. (rambutan), exerts a synergistic effect in the model of human senescent dermal fibroblasts through the same NF- κ B signal inhibition axis. It not only downregulates the expression of core SASP inflammatory factors such as TNF- α and IL-1 β but also enhances the cell's resistance to oxidative stress by activating the Nrf2/ARE antioxidant pathway. Meanwhile, it improves collagen synthesis function and cell migration ability, ultimately reversing the senescent phenotypic characteristics of skin fibroblasts (Lee et al., 2020). The essence of the combined action of these two flavonoids lies in their precise targeting of the NF- κ B signaling hub. By inhibiting the phosphorylation and degradation of I κ B α protein and blocking the interaction between the p65/p50 dimer and the DNA binding domain, they truncate the pro-inflammatory cascade from the transcriptional source. Simultaneously, they synergistically activate the endogenous antioxidant defense system, achieving the

reprogramming of the aging-related inflammatory microenvironment at both the cellular and tissue levels. It is worth noting that the multi-component synergistic effect of rambutan extract (such as the presence of phenolic acids like ellagic acid and gallic acid) may further enhance its anti-inflammatory and anti-aging efficacy in dermal tissues by improving flavonoid bioavailability or providing additional antioxidant support.

- **Metformin:** Metformin synergistically antagonizes cellular senescence through a multi-level network of actions. Its core mechanism starts with the targeted inhibition of mitochondrial complex I (Owen et al., 2000): by blocking the activity of NADH dehydrogenase in the electron transport chain, it effectively reduces superoxide anion leakage, significantly decreases the accumulation of endogenous ROS and the resulting DNA oxidative damage, and alleviates oxidative stress—a key driver of aging—at the source (Algire et al., 2012). In terms of inflammatory regulation, metformin directly interferes with the activation of the IKK kinase complex, inhibits the phosphorylation and degradation of I κ B α protein, and blocks the nuclear translocation of NF- κ B. This disrupts the transcriptional release of pro-inflammatory factors such as IL-6 and TNF- α in the SASP, cutting off the malignant paracrine effects of senescent cells on the microenvironment (Moiseeva et al., 2013). Targeting endoplasmic reticulum stress, a hallmark of aging, metformin specifically upregulates the endoplasmic reticulum-resident glutathione peroxidase 7 (GPX7), enhances the antioxidant capacity of the endoplasmic reticulum lumen, neutralizes hydrogen peroxide produced by the unfolded protein response (UPR), and maintains protein homeostasis (Fang et al., 2018). In a hyperglycemic pathological environment, metformin activates SIRT1 deacetylase, restores the expression levels of key autophagic flux proteins such as LC3B-II and Beclin-1, relieves hyperglycemia-induced autophagy inhibition, and protects lens epithelial cells from premature senescence (Fu et al., 2024). In animal models, short-term metformin intervention can significantly delay radiation-induced premature aging phenotypes. By reducing the expression of p16/p21 senescence markers, it improves muscle function and exercise endurance, confirming its systemic anti-aging efficacy (Fielder et al., 2022).
- **Resveratrol:** Resveratrol and its highly active derivative SRT1720 exert multi-dimensional anti-aging effects by synergistically activating SIRT1 deacetylase. The core mechanism lies in SIRT1-mediated deacetylation of the Lys310 site on the NF- κ B p65 subunit, which directly inhibits NF- κ B transcriptional activity and blocks the release of pro-inflammatory factors such as IL-6 and TNF- α in the SASP. This improves the cell adhesion function of senescent fibroblasts and alleviates the vicious cycle of inflammation (Hubbard and Sinclair, 2014; Pitozzi et al., 2013). In the cardiovascular system, this pathway reduces cellular senescence signals and fibrotic damage after myocardial ischemia by inhibiting the assembly of the NLRP3 inflammasome (Feng et al., 2020). Simultaneously, by activating the Nrf2 antioxidant pathway (Kim et al., 2018a), it synergistically induces the expression of heme oxygenase-1 (HO-1) (Liu et al., 2008), enhances the ability of renal tissues to scavenge free radicals, and delays aging-related glomerulosclerosis. In the field of central nervous system repair, resveratrol drives histone H4K16 deacetylation by restoring the nuclear translocation of SIRT2 protein in senescent OPCs (Ma et al., 2022), initiating the transcriptional expression of myelin regeneration genes (MBP, PLP) and promoting nerve injury repair. In spinal cord injury models, it reduces NF- κ B-dependent neuroinflammatory infiltration of microglia by inhibiting the IKK/I κ B α phosphorylation cascade (Xu et al., 2018). Notably, SRT1720 significantly extends healthspan in both obese mice and mice on a normal diet (Minor et al., 2011; Mitchell et al., 2014). It reverses metabolic aging phenotypes by improving insulin sensitivity and mitochondrial biogenesis. In cervical cancer models, it induces apoptosis of senescent tumor cells through p53 acetylation

modification, revealing a profound intersection between anti-aging and anti-tumor mechanisms (Nadile et al., 2022).

- **Piperlongumine and its analogues:** Piperine and its structurally optimized analogs (compounds 47–49) achieve selective clearance of senescent cells by precisely targeting the interactive hub of oxidative stress and inflammatory pathways. The core mechanism initiates with the specific inhibition of oxidative resistance protein 1 (OXR1), which blocks the peroxidase reduction capacity of this protein, leading to abnormal accumulation of intracellular hydrogen peroxide (H₂O₂) (Zhang et al., 2018). Meanwhile, piperine directly inactivates peroxiredoxin 4 (Prdx4), disrupts the redox balance in the endoplasmic reticulum lumen, and synergistically induces an extreme stress state of the UPR (Kim et al., 2014). This dual oxidative stress forces senescent cells to activate the ATF4-CHOP apoptotic signaling axis, achieving specific clearance of senescent cells by upregulating the pro-apoptotic protein BAX and inducing mitochondrial outer membrane permeabilization (Wang et al., 2016). In terms of neuroimmune regulation, piperine blocks the nuclear translocation of NF- κ B in microglia by interfering with the I κ B α kinase (IKK) phosphorylation cascade, significantly inhibiting the release of neuroinflammatory factors such as TNF- α and IL-1 β , and alleviating aging-related neurodegenerative microenvironmental damage (Kim et al., 2018b). The structurally optimized analogs enhance the stability of I κ B α protein, delay its degradation rate by the ubiquitin-proteasome system, and keep NF- κ B persistently retained in the cytoplasm. This not only strengthens the basic anti-inflammatory effect but also enhances the selective killing efficacy by increasing the oxidative stress threshold in senescent cells (Liu et al., 2018). The core idea of this synergistic approach is to utilize the inherent weakness of senescent cells in coping with oxidative stress. It cleverly uses OXR1/Prdx4 molecules as switches to create a targeted "oxidative storm" in senescent cells while blocking the NF- κ B pathway they use to evade inflammatory attacks. This can efficiently clear senescent cells without harming normal cells, providing a new strategy for achieving precise anti-aging.
- **SR12343:** SR12343, as a new type of small-molecule inhibitor, precisely targets the NEMO protein, a hub node in the IKK/NF- κ B signaling pathway. It mimics the spatial conformation of the NEMO binding domain (NBD) to competitively block the dimerization process of the IKK γ subunit, disrupt the functional assembly of the IKK kinase complex, thereby inhibiting the phosphorylation and degradation of the I κ B α protein, and ultimately blocking the nuclear translocation and transcriptional activation of NF- κ B (Zhao et al., 2018). In terms of aging intervention, this mechanism systematically downregulates the NF- κ B-dependent inflammatory network, significantly reduces the expression of key senescence markers such as p16 and p21, and simultaneously inhibits the release of SASP factors like IL-6 and TNF- α . In premature aging mouse models, it effectively improves multi-organ degenerative phenotypes such as muscle function decline and cognitive impairment, and increases the average healthy lifespan (Zhang et al., 2021). Different from traditional IKK catalytic activity inhibitors, SR12343 allosterically interferes with the protein interaction interface of the regulatory subunit NEMO, blocks the activation of the IKK complex with higher selectivity, and greatly reduces the risk of off-target toxicity. Meanwhile, short-term intervention can reverse radiation-induced aging-related frailty syndrome, confirming its translational potential as an efficient aging intervention strategy.
- **Curcumin and its metabolite:** Curcumin and its main metabolite vanillin achieve selective clearance of senescent cells by synergistically regulating the Nrf2 antioxidant pathway and NF- κ B inflammatory signals. The core mechanism starts with curcumin activating the Nrf2 transcription factor, which significantly upregulates the expression of HO-1 by binding to the antioxidant response element (ARE), thereby enhancing the intracellular endogenous antioxidant capacity to neutralize the accumulation of reactive oxygen species

related to aging (Balogun et al., 2003). Meanwhile, vanillin interferes with the histone modification network, promotes the binding of HDAC3 to the MafK transcription complex, and blocks the co-activation of Nrf2 by the CREB-binding protein (CBP), forming an interactive inhibition of Nrf2/NF- κ B signals (Cherif et al., 2019). This dual intervention triggers a synergistic effect in the model of human intervertebral disc senescent cells: on the one hand, it blocks the nuclear translocation of NF- κ B by inhibiting the degradation of I κ B α , and downregulates the release of SASP pro-inflammatory factors such as IL-6 and MMP-3; on the other hand, it increases the ratio of pro-apoptotic protein BAX/BCL-2 and activates the caspase-3-dependent apoptotic pathway to selectively clear senescent cells. It is worth noting that as an active metabolite of curcumin in vivo, vanillin, with its small molecular property, enhances the permeability to intervertebral disc tissues. By targeting the epigenetic reprogramming weaknesses specific to senescent cells, especially mechanisms such as abnormal recruitment of HDAC3, it achieves precise repair of the microenvironment of degenerated intervertebral discs.

- **Others:** Endogenous glucocorticoid cortisol precisely truncates the positive feedback loop of inflammatory signals (Laberge et al., 2012). The core of its mechanism lies in the competitive binding of glucocorticoid receptor (GR) to the promoter region of the IL-1 α gene, which directly inhibits the transcriptional activation of IL-1 α , thereby disrupting the IL-1 α /NF- κ B autocrine amplifying inflammatory cycle. This targeted intervention selectively blocks the release of tissue-destructive factors such as MMPs and IL-8 in SASP without significantly affecting other physiological inflammatory mediators, achieving precise inhibition of the paracrine toxicity of senescent cells. Oleuropein, an olive polyphenol, and its hydrolyzed metabolite hydroxytyrosol synergistically activate the PPAR γ nuclear receptor, induce direct protein interaction between PPAR γ and the NF- κ B p65 subunit, form a transcriptional inhibitory complex, block the ubiquitination and degradation process of I κ B α , and keep NF- κ B persistently retained in the cytoplasm (Menicacci et al., 2017). Meanwhile, as a potent antioxidant, hydroxytyrosol indirectly inhibits ROS-dependent IKK kinase activation by quenching ROS, and downregulates the expression of core SASP inflammatory factors such as TNF- α and IL-1 β through this dual-track approach. In the model of human senescent fibroblasts, the olive polyphenol mixture, through the synergy between the PPAR γ -I κ B α axis and the antioxidant pathway, significantly improves cell migration ability and collagen synthesis function, and reverses the senescent phenotype of dermal tissues.

5.2.3. p38MAPK inhibitors

Inhibitors of the p38MAPK pathway (SB203580, UR13756, BIRB796) and inhibitors of its downstream effector kinase MK2 (PF-3644022, MK2.III) synergistically block the core signal transduction of SASP by targeting the DNA damage-independent regulatory hub of SASP (Freund et al., 2011). The core of their mechanism lies in inhibiting the phosphorylation and activation of MK2 kinase by p38MAPK, disrupting the MK2-mediated mRNA stability regulatory network. This leads to the inactivation of binding proteins in the 3'-UTR region of mRNAs encoding key SASP factors such as IL-6 and IL-8, accelerating the degradation of transcripts of these pro-inflammatory factors (Alimbetov et al., 2016). Meanwhile, this pathway amplifies IL-1 α signals through positive feedback. Inhibiting p38MAPK activity can directly reduce IL-1 α protein synthesis, truncate the IL-1 α /NF- κ B autocrine loop, and synergistically block NF- κ B nuclear translocation as well as the expression of downstream tissue-destructive factors such as MMP-3 and GRO α (Freund et al., 2011). In the model of human senescent fibroblasts, the combination of BIRB796 (a highly selective p38 inhibitor) and PF-3644022 (an MK2 allosteric inhibitor) produces a superadditive effect. By simultaneously interfering with the p38-MK2 signaling axis and the p38-TTP (tristetraprolin mRNA degradation factor) interaction network, the

secretion level of SASP factors is reduced by more than 80 %, and the cell migration and extracellular matrix remodeling abilities are significantly restored (Alimbetov et al., 2016). Compared with DNA damage repair-targeting drugs, p38MAPK/MK2 inhibitors directly intervene in the unique inflammatory signal amplification mechanism of senescent cells. Through the dual approaches of blocking the IL-1 α positive feedback loop and regulating mRNA stability, they achieve the source containment of SASP storms, providing a precise intervention window for delaying tissue degeneration.

5.2.4. JAK inhibitors

JAK inhibitors (ruxolitinib and momelotinib) exert anti-aging effects in multiple tissues by precisely targeting the JAK-STAT inflammatory signaling axis. The core mechanism lies in blocking the phosphorylation activity of JAK1/2 kinases, inhibiting the tyrosine 705 site phosphorylation and nuclear translocation of STAT3 transcription factor, and directly downregulating the gene transcription of SASP factors such as IL-6 and MMP-3. This alleviates systemic chronic inflammatory states (Xu et al., 2015b), significantly improves frailty syndrome in elderly individuals, and enhances exercise endurance (Griveau et al., 2020). In the skeletal system, ruxolitinib reverses the senescence of the bone marrow microenvironment, blocks the JAK2/STAT3-mediated attenuation of estrogen receptor α (ER α) signaling, restores the osteogenic differentiation ability of bone marrow mesenchymal stem cells, and inhibits the excessive activation of osteoclasts (Wu et al., 2020). In elderly mouse models, it effectively increases bone mineral density and reduces the risk of fractures (Farr et al., 2017). For metabolic aging, JAK inhibitors activate the STAT5-PPAR γ signaling axis in adipose tissue, promote the differentiation of senescent adipose stem cells into functional adipocytes, improve adiponectin secretion and insulin sensitivity, and reverse age-related metabolic dysfunction (Xu et al., 2015a). Particularly, the clinical application of ruxolitinib in patients with myelofibrosis has confirmed that it can reduce pathological inflammation in the bone marrow microenvironment, indirectly suggesting its potential to repair aging-related hematopoietic stem cell exhaustion (Harrison et al., 2016).

5.2.5. ATM kinase inhibitors

ATM kinase inhibitors (KU-60019 and KU-55933) exert anti-aging effects in multiple tissues by truncating the vicious cycle between DDR and inflammatory aging. The core mechanism lies in blocking the persistent signal amplification of ATM kinase in response to DNA DSBs (Kang et al., 2017). By inhibiting the ATM-p38MAPK positive feedback loop, they significantly reduce the expression of senescence markers such as p16 and p21 (Kuk et al., 2019). Meanwhile, this intervention selectively relieves autophagy inhibition: by restoring the autophagic degradation of GATA4 transcription factor, it prevents the accumulation of GATA4 that would otherwise activate the NF- κ B pathway, thus cutting off the transformation process from DNA damage to SASP inflammatory phenotype at the source (Kang et al., 2015). In accelerated aging models, KU-60019 remodels the stem cell microenvironment, inhibits ATM-mediated nuclear translocation and transcriptional activation of NF- κ B, alleviates the toxic pressure of inflammatory factors TNF- α /IL-1 β on hematopoietic stem cells and NSCs, and improves the self-renewal and differentiation abilities of stem cells (Zhao et al., 2020). It is worth noting that in premature aging mouse models, short-term intervention with ATM inhibitors can significantly delay multi-organ degenerative changes such as osteoporosis and muscle atrophy, confirming the systemic anti-aging potential of its regulatory mechanism (Kuk et al., 2019).

6. Discussion

Cellular senescence acts as a central mechanism in the pathogenesis of AD, inducing irreversible cell cycle arrest in neurons, glial cells, and cerebrovascular endothelial cells through the synergistic activation of

the p53/p21 and p16/RB signaling pathways. Senescent cells also secrete SASP factors that directly exacerbate A β deposition, tau hyperphosphorylation-induced neurotoxicity, neuroinflammatory cascades, BBB dysfunction, and mitochondrial energy deficits, thereby establishing a cross-cell-type "senescence-neurodegeneration" feedback loop. Targeting this pathological cascade, senotherapeutics have demonstrated significant therapeutic potential. Specifically, senolytics selectively eliminate senescent cells by exploiting their unique survival dependencies—such as reliance on anti-apoptotic pathways, elevated lysosomal β -gal activity, and mitochondrial metabolic impairments—resulting in reduced tau phosphorylation and A β accumulation, restored synaptic plasticity, and improved cognitive function in pre-clinical models. Early-phase clinical trials further support these findings, showing reductions in senescence biomarkers in peripheral blood and cerebrospinal fluid alongside improvements in cognitive performance and activities of daily living. In contrast, senomorphics mitigate neuroinflammation and delay microenvironmental deterioration by inhibiting key SASP-regulating pathways, including mTOR, NF- κ B, p38MAPK, and JAK-STAT. Nevertheless, several challenges remain: variable efficacy due to the spatiotemporal heterogeneity of senescent cells in the brain, off-target toxicity of senolytics to non-senescent tissues (e.g., platelets), limited blood-brain barrier penetration, and insufficient data on long-term safety.

Notably, a central controversy in the field remains unresolved: whether post-mitotic neurons fully meet the classical definition of cellular senescence, which is traditionally defined by replicative arrest. Saez-Atienzar et al. (Saez-Atienzar and Masliah, 2020) argued that neuronal "senescence-like phenotypes," such as DNA damage and lipofuscin accumulation, may reflect a distinct, AD-specific degenerative state rather than canonical senescence. This distinction is crucial for identifying cell-type-specific therapeutic targets. Beyond the debate over neuronal senescence's classical definition, another notable unresolved issue concerns the cell type-specific predominance of p16- versus p21-mediated senescence pathways. In the murine hippocampus, p16-positive cells are primarily microglia and oligodendrocytes, exhibit a marked age-dependent increase across all subpopulations, and display high heterogeneity in SASP factor expression. Conversely, p21-positive cells are predominantly microglial, show age-related increases only within specific clusters, and express a more uniform and conserved SASP profile with minimal overlap with p16-positive populations. Moreover, these two senescent cell populations follow distinct developmental trajectories, with no evidence of interconversion. Even among the same cell types, their SASP profiles differ significantly, suggesting that these differences are driven by p16 and p21 expression rather than intrinsic cellular identity (Saul et al., 2025). This cell type-specific divergence in the murine hippocampus reflects broader patterns of context-dependent senescence heterogeneity, underscoring the complexity of cellular senescence and SASP regulation across tissues and offering crucial insights for understanding aging mechanisms and refining targeted senolytic interventions.

7. Conclusion

To address the unresolved controversies, inherent challenges, and cell-type-specific heterogeneity of senescence in AD—including the ambiguous definition of neuronal senescence, divergent roles of p16- versus p21-mediated pathways, spatiotemporal variability of senescent cells, off-target toxicity of senolytics, and limited blood-brain barrier penetration—future research should prioritize targeted and translational advances. Specifically, efforts should focus on developing spatiotemporally precise delivery systems (e.g., uPAR-CAR-T cells, SA- β -gal-activated prodrugs, and brain-targeted nanocarriers) to enhance the specificity and bioavailability of senotherapeutics. Simultaneously, elucidating cell type-specific molecular signatures of senescence in the AD brain will refine targeted interventions, while evaluating the synergistic effects of combined senolytic and senomorphic therapies may

overcome the limitations of single-agent strategies. Additionally, integrating multi-omics approaches and organoid-based models will help map the dynamic crosstalk between senescent cells and AD pathology, providing mechanistic insights to guide clinical translation. Collectively, these coordinated efforts will bridge the gap between preclinical discoveries and clinical application of senotherapeutics, paving the way for a novel disease-modifying paradigm not only for AD but also for other age-related neurodegenerative disorders.

Ethics approval

The Research Ethics Committee has confirmed that no ethical approval is required.

Funding

This work was supported by the University-Hospital Joint Fund Project of Guangzhou University of Chinese Medicine (GZYDG2024Y05 to Runjin Zhou, GZYDG2024Y02 to Haojie Ni). National Key R&D Program of China (2022YFC3501400 to Xingdong Lin). The Dongguan Social Development Technology Project (Special Projects for High-level Hospital Construction) (20231800915372 to Weimin Ning). The "Inheriting Studio of Provincial Renowned Chinese Medicine Practitioner Ning Weimin" of the Provincial Administration of Traditional Chinese Medicine (Yue Zhongyi Ban Han [2023] No. 108 to Weimin Ning). The "Construction Project of Dongguan City Renowned Chinese Medicine Practitioner Ning Weimin's Inheriting Studio" of Dongguan Health Bureau (Dong Wei Ban [2019] No. 36 to Weimin Ning). National Natural Science Foundation of China (NSFC) (82374571 and 82074042 to Juxian Song, 82575202, 82174479 to Chunzhi Tang, and 82305130 to Qiuxing He).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Not applicable.

Authors' contributions

Xingdong Lin, Chunzhi Tang, Juxian Song, and Weimin Ning conceived and designed the experiments. Material preparation, data collection, and analysis were performed by Runjin Zhou, Xiaoling Lin, and Zheng Liao. Runjin Zhou, Jian Lin, Ruixue Zheng, Haojie Ni, Wei Liu, Zhanpeng Feng, and Qiuxing He prepared figures and tables. Runjin Zhou and Xiaoling Lin drafted the manuscript, Juxian Song and Weimin Ning reviewed and edited the final draft. All authors reviewed and approved the final manuscript.

Clinical trial number

Not applicable.

Availability of data and materials

All data supporting the findings of this study are presented in the main text and accompanying figures and tables.

References

- Abdelgawad, I.Y., Agostinucci, K., Ismail, S.G., Grant, M.K.O., Zordoky, B.N., 2022. EA hy926 cells and HUVECs share similar senescence phenotypes but respond differently to the senolytic drug ABT-263. *Cells* 11 (13).
- Abokyi, S., Ghartey-Kwansah, G., Tse, D.Y.-Y., 2023. TFEB is a central regulator of the aging process and age-related diseases. *Ageing Res Rev.* 89, 101985.
- Ahmad, A., Braden, A., Khan, S., Xiao, J., Khan, M.M., 2024. Crosstalk between the DNA damage response and cellular senescence drives aging and age-related diseases. *Semin Immunopathol.* 46 (3-4), 10.
- Algire, C., Moiseeva, O., Deschênes-Simard, X., Amrein, L., Petrucci, L., Birman, E., Viollet, B., Ferbeyre, G., Pollak, M.N., 2012. Metformin reduces endogenous reactive oxygen species and associated DNA damage. *Cancer Prev Res (Phila)* 5 (4), 536–543.
- Alimbetov, D., Davis, T., Brook, A.J.C., Cox, L.S., Faragher, R.G.A., Nurgozhin, T., Zhumadilov, Z., Kipling, D., 2016. Suppression of the senescence-associated secretory phenotype (SASP) in human fibroblasts using small molecule inhibitors of p38 MAP kinase and MK2. *Biogerontology* 17 (2), 305–315.
- Al-Mashhadi, S., Simpson, J.E., Heath, P.R., Dickman, M., Forster, G., Matthews, F.E., Brayne, C., Ince, P.G., Wharton, S.B., 2015. Oxidative glial cell damage associated with white matter lesions in the aging human brain. *Brain Pathol.* 25 (5), 565–574.
- Amor, C., Feucht, J., Leibold, J., Ho, Y.-J., Zhu, C., Alonso-Curbelo, D., Mansilla-Soto, J., Boyer, J.A., Li, X., Giavridis, T., Kulick, A., Houlihan, S., Peerschke, E., Friedman, S. L., Ponomarev, V., Piersigilli, A., Sadelain, M., Lowe, S.W., 2020. Senolytic CAR T cells reverse senescence-associated pathologies. *Nature* 583 (7814), 127–132.
- Anderson, R., Lagnado, A., Maggiorani, D., Walaszczyk, A., Dookun, E., Chapman, J., Birch, J., Salmonowicz, H., Ogorodnik, M., Jurk, D., Proctor, C., Correia-Melo, C., Vitorcelli, S., Fielder, E., Berlinguer-Palmini, R., Owens, A., Greaves, L.C., Kolsky, K. L., Parini, A., Douin-Echinard, V., LeBrasseur, N.K., Arthur, H.M., Adam-Chalot, S., Schaefer, M.J., Roos, C.M., Miller, J.D., Robertson, N., Mann, J., Adams, P.D., Tchkonja, T., Kirkland, J.L., Miallet-Perez, J., Richardson, G.D., Passos, J.F., 2019. Length-independent telomere damage drives post-mitotic cardiomyocyte senescence. *EMBO J.* 38 (5).
- Anisimov, V.N., Zabezhinski, M.A., Popovich, I.G., Piskunova, T.S., Semchenko, A.V., Tyndyk, M.L., Yurova, M.N., Antoch, M.P., Blagosklonny, M.V., 2010. Rapamycin extends maximal lifespan in cancer-prone mice. *Am. J. Pathol.* 176 (5), 2092–2097.
- Baar, M.P., Brandt, R.M.C., Putavet, D.A., Klein, J.D.D., Derks, K.W.J., Bourgeois, B.R.M., Stryck, S., Rijkse, Y., van Willigenburg, H., Feijtel, D.A., van der Pluijm, I., Essers, J., van Cappellen, W., van Ijcken, W.F., Houtsmuller, A.B., Pothof, J., de Bruin, R.W.F., Madl, T., Hoeijmakers, J.H.J., Campisi, J., de Keizer, P.L.J., 2017. Targeted apoptosis of senescent cells restores tissue homeostasis in response to chemotoxicity and aging. *Cell* 169 (1).
- Bachstetter, A.D., Xing, B., de Almeida, L., Dimayuga, E.R., Watterson, D.M., Van Eldik, L.J., 2011. Microglial p38 α MAPK is a key regulator of proinflammatory cytokine up-regulation induced by toll-like receptor (TLR) ligands or beta-amyloid (A β). *J. Neuroinflamm.* 8, 79.
- Baker, D.J., Perez-Terzic, C., Jin, F., Pitel, K.S., Niederländer, N.J., Jeganathan, K., Yamada, S., Reyes, S., Rowe, L., Hiddinga, H.J., Eberhardt, N.L., Terzic, A., van Deursen, J.M., 2008. Opposing roles for p16lnk4a and p19Arf in senescence and ageing caused by BubR1 insufficiency. *Nat. Cell Biol.* 10 (7), 825–836.
- Balogun, E., Hoque, M., Gong, P., Killeen, E., Green, C.J., Foresti, R., Alam, J., Motterlini, R., 2003. Curcumin activates the haem oxygenase-1 gene via regulation of Nrf2 and the antioxidant-responsive element. *Biochem J.* 371 (Pt 3), 887–895.
- Bhat, R., Crowe, E.P., Bitto, A., Moh, M., Katsetos, C.D., Garcia, F.U., Johnson, F.B., Trojanowski, J.Q., Sell, C., Torres, C., 2012. Astrocyte senescence as a component of Alzheimer's disease. *PLoS One* 7 (9), e45069.
- Bitto, A., Sell, C., Crowe, E., Lorenzini, A., Malaguti, M., Hrelia, S., Torres, C., 2010. Stress-induced senescence in human and rodent astrocytes. *Exp. Cell Res* 316 (17), 2961–2968.
- Bjedov, I., Rallis, C., 2020. The target of rapamycin signalling pathway in ageing and lifespan regulation. *Genes (Basel)* 11 (9).
- Bose, R., Moors, M., Tofighi, R., Cascante, A., Hermanson, O., Ceccatelli, S., 2010. Glucocorticoids induce long-lasting effects in neural stem cells resulting in senescence-related alterations. *Cell Death Dis.* 1 (11), e92.
- Brickman, A.M., 2013. Contemplating Alzheimer's disease and the contribution of white matter hyperintensities. *Curr. Neurol. Neurosci. Rep.* 13 (12), 415.
- Buscemi, G., Perego, P., Carenini, N., Nakanishi, M., Chessa, L., Chen, J., Khanna, K., Delia, D., 2004. Activation of ATM and Chk2 kinases in relation to the amount of DNA strand breaks. *Oncogene* 23 (46), 7691–7700.
- Bussian, T.J., Aziz, A., Meyer, C.F., Swenson, B.L., van Deursen, J.M., Baker, D.J., 2018. Clearance of senescent glial cells prevents tau-dependent pathology and cognitive decline. *Nature* 562 (7728), 578–582.
- Cai, Y., Zhou, H., Zhu, Y., Sun, Q., Ji, Y., Xue, A., Wang, Y., Chen, W., Yu, X., Wang, L., Chen, H., Li, C., Luo, T., Deng, H., 2020. Elimination of senescent cells by β -galactosidase-targeted prodrug attenuates inflammation and restores physical function in aged mice. *Cell Res* 30 (7), 574–589.
- Carlos, A.J., Tong, L., Prieto, G.A., Cotman, C.W., 2017. IL-1 β impairs retrograde flow of BDNF signaling by attenuating endosome trafficking. *J. Neuroinflamm.* 14 (1), 29.
- Chang, J., Wang, Y., Shao, L., Laberge, R.-M., Demaria, M., Campisi, J., Janakiraman, K., Sharpless, N.E., Ding, S., Feng, W., Luo, Y., Wang, X., Aykin-Burns, N., Krager, K., Ponnappan, U., Hauer-Jensen, M., Meng, A., Zhou, D., 2016. Clearance of senescent cells by ABT263 rejuvenates aged hematopoietic stem cells in mice. *Nat. Med* 22 (1), 78–83.
- Chauhan, D., Tian, Z., Nicholson, B., Kumar, K.G.S., Zhou, B., Carrasco, R., McDermott, J. L., Leach, C.A., Fulciniti, M., Kodrasov, M.P., Weinstock, J., Kingsbury, W.D., Hideshima, T., Shah, P.K., Minvielle, S., Altun, M., Kessler, B.M., Orlowski, R., Richardson, P., Munshi, N., Anderson, K.C., 2012. A small molecule inhibitor of ubiquitin-specific protease-7 induces apoptosis in multiple myeloma cells and overcomes bortezomib resistance. *Cancer Cell* 22 (3), 345–358.
- Chen, Q., Gu, P., Liu, X., Hu, S., Zheng, H., Liu, T., Li, C., 2022. Gold nanoparticles encapsulated resveratrol as an anti-aging agent to delay cataract development. *Pharm. (Basel)* 16 (1).
- Cheng, T., Grasse, L., Shah, J., Chandra, J., 2015. Panobinostat, a pan-histone deacetylase inhibitor: rationale for and application to treatment of multiple myeloma. *Drugs Today (Barc)* 51 (8), 491–504.
- Cherif, H., Bisson, D.G., Jarzem, P., Weber, M., Ouellet, J.A., Haglund, L., 2019. Curcumin and o-vanillin exhibit evidence of senolytic activity in human IVD cells in vitro. *J. Clin. Med* 8 (4).
- Cherif, H., Bisson, D.G., Mannarino, M., Rabau, O., Ouellet, J.A., Haglund, L., 2020. Senotherapeutic drugs for human intervertebral disc degeneration and low back pain. *Elife* 9.
- Childs, B.G., Baker, D.J., Kirkland, J.L., Campisi, J., van Deursen, J.M., 2014. Senescence and apoptosis: dueling or complementary cell fates? *EMBO Rep.* 15 (11), 1139–1153.
- Childs, B.G., Baker, D.J., Wijshake, T., Conover, C.A., Campisi, J., van Deursen, J.M., 2016. Senescent intimal foam cells are deleterious at all stages of atherosclerosis. *Science* 354 (6311), 472–477.
- Chinta, S.J., Woods, G., Demaria, M., Rane, A., Zou, Y., McQuade, A., Rajagopalan, S., Limbad, C., Madden, D.T., Campisi, J., Andersen, J.K., 2018. Cellular senescence is induced by the environmental neurotoxin paraquat and contributes to neuropathology linked to Parkinson's disease. *Cell Rep.* 22 (4), 930–940.
- Clarke, L.E., Liddelow, S.A., Chakraborty, C., Münch, A.E., Heiman, M., Barres, B.A., 2018. Normal aging induces A1-like astrocyte reactivity. *Proc. Natl. Acad. Sci. USA* 115 (8), E1896–E1905.
- Conde, J.R., Streit, W.J., 2006. Microglia in the aging brain. *J. Neuropathol. Exp. Neurol.* 65 (3), 199–203.
- Coppé, J.-P., Patil, C.K., Rodier, F., Sun, Y., Muñoz, D.P., Goldstein, J., Nelson, P.S., Desprez, P.-Y., Campisi, J., 2008. Senescence-associated secretory phenotypes reveal cell-nonautonomous functions of oncogenic RAS and the p53 tumor suppressor. *PLoS Biol.* 6 (12), 2853–2868.
- Crowe, E.P., Tuzer, F., Gregory, B.D., Donahue, G., Gosai, S.J., Cohen, J., Leung, Y.Y., Yetkin, E., Nativio, R., Wang, L.-S., Sell, C., Bonini, N.M., Berger, S.L., Johnson, F.B., Torres, C., 2016. Changes in the transcriptome of human astrocytes accompanying oxidative stress-induced senescence. *Front. Aging Neurosci.* 8, 208.
- Currais, A., Farrokhi, C., Dargusch, R., Armando, A., Quehenberger, O., Schubert, D., Maher, P., 2018. Fisetin reduces the impact of aging on behavior and physiology in the rapidly aging SAMP8 mouse. *J. Gerontol. A Biol. Sci. Med. Sci.* 73 (3), 299–307.
- d'Adda di Fagnagna, F., 2008. Living on a break: cellular senescence as a DNA-damage response. *Nat. Rev. Cancer* 8 (7), 512–522.
- D'Alessandro, M.C.B., Kanaan, S., Geller, M., Praticò, D., Daher, J.P.L., 2025. Mitochondrial dysfunction in Alzheimer's disease. *Ageing Res Rev.* 107, 102713.
- Daly, T., Kepp, K.P., Imbimbo, B.P., 2024. Are lecanemab and donanemab disease-modifying therapies? *Alzheimers Dement* 20 (9), 6659–6661.
- Daniele, S., Da Pozzo, E., Iofrida, C., Martini, C., 2016. Human neural stem cell aging is counteracted by α -glycerylphosphorylethanolamine. *ACS Chem. Neurosci.* 7 (7), 952–963.
- Dawson, M.R.L., Politò, A., Levine, J.M., Reynolds, R., 2003. NG2-expressing glial progenitor cells: an abundant and widespread population of cycling cells in the adult rat CNS. *Mol. Cell Neurosci.* 24 (2), 476–488.
- Dehkhordi, S.K., Walker, J., Sah, E., Bennett, E., Atrian, F., Frost, B., Woost, B., Bennett, R. E., Orr, T.C., Zhou, Y., Andhey, P.S., Colonna, M., Sudmant, P.H., Xu, P., Wang, M., Zhang, B., Zare, H., Orr, M.E., 2021. Profiling senescent cells in human brains reveals neurons with CDKN2D/p19 and tau neuropathology. *Nat. Aging* 1 (12), 1107–1116.
- Del Gaudio Moore, V., Brown, J.R., Certo, M., Love, T.M., Novina, C.D., Letai, A., 2007. Chronic lymphocytic leukemia requires BCL2 to sequester prodeath BIM, explaining sensitivity to BCL2 antagonist ABT-737. *J. Clin. Invest* 117 (1), 112–121.
- van Delft, M.F., Wei, A.H., Mason, K.D., Vandenberg, C.J., Chen, L., Czabotar, P.E., Willis, S.N., Scott, C.L., Day, C.L., Cory, S., Adams, J.M., Roberts, A.W., Huang, D.C. S., 2006. The BH3 mimetic ABT-737 targets selective Bcl-2 proteins and efficiently induces apoptosis via Bak/Bax if Mcl-1 is neutralized. *Cancer Cell* 10 (5), 389–399.
- Demaria, M., O'Leary, M.N., Chang, J., Shao, L., Liu, S., Alimirah, F., Koenig, K., Le, C., Mitin, N., Deal, A.M., Alston, S., Academia, E.C., Kilmarx, S., Valdivinos, A., Wang, B., de Bruin, A., Kennedy, B.K., Melov, S., Zhou, D., Sharpless, N.E., Muss, H., Campisi, J., 2017. Cellular senescence promotes adverse effects of chemotherapy and cancer relapse. *Cancer Discov.* 7 (2), 165–176.
- Doan, L., Paine, P., Tran, C., Parsons, B., Hiller, A., Joshua, I., Collaco, N., Vitari, A., Antoine, D.M., Ishak, H., Jasper, J., Gallop, M., Luo, J., Tony, W.-C., Rando, T., Quarta, M., 2020. Targeted senolytic prodrug is well tolerated and results in amelioration of frailty, muscle regeneration and cognitive functions in geriatric mice. *Research Square*. (<https://www.researchsquare.com/article/rs-92962/v1>).
- van Dyck, C.H., Swanson, C.J., Aisen, P., Bateman, R.J., Chen, C., Gee, M., Kanekiyo, M., Li, D., Reyderman, L., Cohen, S., Froelich, L., Katayama, S., Sabbagh, M., Vellas, B., Watson, D., Dhadda, S., Irizarry, M., Kramer, L.D., Iwatsubo, T., 2023. Lecanemab in early Alzheimer's disease. *N. Engl. J. Med* 388 (1).
- Edwards, A.L., Collins, J.A., Junge, C., Kordasiewicz, H., Mignon, L., Wu, S., Li, Y., Lin, L., DuBois, J., Hutchison, R.M., Ziogas, N., Shulman, M., Martarello, L., Graham, D., Lane, R., Budd Haerlein, S., Beaver, J., 2023. Exploratory tau biomarker results from a multiple ascending-dose study of BILB080 in Alzheimer disease: a randomized clinical trial. *JAMA Neurol.* 80 (12), 1344–1352.
- Eftekharpour, E., 2023. The neuronal nucleus: a new battlefield in fight against neurodegeneration. *Aging (Albany NY)* 15 (4), 898–904.

- Erten-Lyons, D., Woltjer, R., Kaye, J., Mattek, N., Dodge, H.H., Green, S., Tran, H., Howieson, D.B., Wild, K., Silbert, L.C., 2013. Neuropathologic basis of white matter hyperintensity accumulation with advanced age. *Neurology* 81 (11), 977–983.
- Evans, R.J., Wyllie, F.S., Wynford-Thomas, D., Kipling, D., Jones, C.J., 2003. A P53-dependent, telomere-independent proliferative life span barrier in human astrocytes consistent with the molecular genetics of glioma development. *Cancer Res* 63 (16), 4854–4861.
- Fang, J., Yang, J., Wu, X., Zhang, G., Li, T., Wang, X., Zhang, H., Wang, C.-C., Liu, G.-H., Wang, L., 2018. Metformin alleviates human cellular aging by upregulating the endoplasmic reticulum glutathione peroxidase 7. *Aging Cell* 17 (4), e12765.
- Farina, C., Aloisi, F., Meinel, E., 2007. Astrocytes are active players in cerebral innate immunity. *Trends Immunol.* 28 (3), 138–145.
- Farr, J.N., Xu, M., Weivoda, M.M., Monroe, D.G., Fraser, D.G., Onken, J.L., Negley, B.A., Sfeir, J.G., Ogronnik, M.B., Hachfeld, C.M., LeBrasseur, N.K., Drake, M.T., Pignolo, R.J., Pirtskhalava, T., Tchonia, T., Oursler, M.J., Kirkland, J.L., Khosla, S., 2017. Targeting cellular senescence prevents age-related bone loss in mice. *Nat. Med* 23 (9), 1072–1079.
- Feng, H., Mou, S.-Q., Li, W.-J., Zhang, N., Zhou, Z.-Y., Ding, W., Bian, Z.-Y., Liao, H.-H., 2020. Resveratrol inhibits ischemia-induced myocardial senescence signals and nlrp3 inflammasome activation. *Oxid. Med Cell Longev.* 2020, 2647807.
- Ferrón, S., Mira, H., Franco, S., Cano-Jaimez, M., Bellmunt, E., Ramírez, C., Fariñas, I., Blasco, M.A., 2004. Telomere shortening and chromosomal instability abrogates proliferation of adult but not embryonic neural stem cells. *Development* 131 (16), 4059–4070.
- Ferrón, S.R., Marqués-Torrejón, M.A., Mira, H., Flores, I., Taylor, K., Blasco, M.A., Fariñas, I., 2009. Telomere shortening in neural stem cells disrupts neuronal differentiation and neurogenesis. *J. Neurosci.* 29 (46), 14394–14407.
- Fielder, E., Wan, T., Alimohammadi, G., Ishaq, A., Low, E., Weigand, B.M., Kelly, G., Parker, C., Griffin, B., Jurk, D., Korolchuk, V.I., von Zglinicki, T., Miwa, S., 2022. Short senolytic or senostatic interventions rescue progression of radiation-induced frailty and premature ageing in mice. *Elife* 11.
- Flanary, B.E., Sammons, N.W., Nguyen, C., Walker, D., Streit, W.J., 2007. Evidence that aging and amyloid promote microglial cell senescence. *Rejuvenation Res* 10 (1), 61–74.
- Forero, D.A., González-Giraldo, Y., López-Quintero, C., Castro-Vega, L.J., Barreto, G.E., Perry, G., 2016b. Telomere length in Parkinson's disease: a meta-analysis. *Exp. Gerontol.* 75, 53–55.
- Forero, D.A., González-Giraldo, Y., López-Quintero, C., Castro-Vega, L.J., Barreto, G.E., Perry, G., 2016a. Meta-analysis of telomere length in Alzheimer's disease. *J. Gerontol. A Biol. Sci. Med Sci.* 71 (8), 1069–1073.
- Freund, A., Patil, C.K., Campisi, J., 2011. p38MAPK is a novel DNA damage response-independent regulator of the senescence-associated secretory phenotype. *EMBO J.* 30 (8), 1536–1548.
- Frost, B., Bardai, F.H., Feany, M.B., 2016. Lamin dysfunction mediates neurodegeneration in tauopathies. *Curr. Biol.* 26 (1), 129–136.
- Fu, Y., Wu, R., Dong, S., Chen, J., Zhou, N., 2024. Metformin protects human lens epithelial cells from high glucose-induced senescence and autophagy inhibition by upregulating SIRT1. *Graefes Arch. Clin. Exp. Ophthalmol.* 262 (2), 477–485.
- Fuhrmann-Stroissnigg, H., Ling, Y.Y., Zhao, J., McGowan, S.J., Zhu, Y., Brooks, R.W., Grassi, D., Gregg, S.Q., Stripay, J.L., Dorronsoro, A., Corbo, L., Tang, P., Bukata, C., Ring, N., Giacca, M., Li, X., Tchonia, T., Kirkland, J.L., Niedernhofer, L.J., Robbins, P.D., 2017. Identification of HSP90 inhibitors as a novel class of senolytics. *Nat. Commun.* 8 (1), 422.
- Gage, F.H., 2000. Mammalian neural stem cells. *Science* 287 (5457), 1433–1438.
- Gaikwad, S., Puangmalai, N., Bittar, A., Montalbano, M., Garcia, S., McAllen, S., Bhatt, N., Sonawane, M., Sengupta, U., Kaye, R., 2021. Tau oligomer induced HMGB1 release contributes to cellular senescence and neuropathology linked to Alzheimer's disease and frontotemporal dementia. *Cell Rep.* 36 (3), 109419.
- Garbarino, V.R., Palavicini, J.P., Melendez, J., Barthelemy, N.R., He, Y., Kautz, T.F., Lopez-Cruzan, M., Mathews, J.J., Xu, P., Zhang, B., Saliba, A., Ragi, N., Sharma, K., Mason, D., Johnson, S., Hendrix, S., Craft, S., Petersen, R.C., Espindola-Netto, J.M., Xue, A., Tchonia, T., Kirkland, J.L., Salardini, A., Musi, N., Bateman, R.J., Gonzales, M.M., Orr, M.E., 2025. Evaluation of exploratory fluid biomarkers from a phase 1 senolytic trial in mild Alzheimer's disease. *Neurotherapeutics*, e00591.
- Geng, Y.-Q., Guan, J.-T., Xu, X.-H., Fu, Y.-C., 2010. Senescence-associated beta-galactosidase activity expression in aging hippocampal neurons. *Biochem Biophys. Res Commun.* 396 (4), 866–869.
- Giacchi, M.R., Bartlett, C.A., Smith, N.M., Iyer, K.S., Toomey, L.M., Jiang, H., Guagliardo, P., Kilburn, M.R., Fitzgerald, M., 2018. Oligodendroglia are particularly vulnerable to oxidative damage after neurotrauma in vivo. *J. Neurosci.* 38 (29), 6491–6504.
- Gong, L., Gong, H., Pan, X., Chang, C., Ou, Z., Ye, S., Yin, L., Yang, L., Tao, T., Zhang, Z., Liu, C., Lane, D.P., Peng, J., Chen, J., 2015. p53 isoform $\Delta 113p53/\Delta 133p53$ promotes DNA double-strand break repair to protect cell from death and senescence in response to DNA damage. *Cell Res* 25 (3), 351–369.
- Gonzales, M.M., Garbarino, V.R., Kautz, T.F., Palavicini, J.P., Lopez-Cruzan, M., Dehkordi, S.K., Mathews, J.J., Zare, H., Xu, P., Zhang, B., Franklin, C., Habes, M., Craft, S., Petersen, R.C., Tchonia, T., Kirkland, J.L., Salardini, A., Seshadri, S., Musi, N., Orr, M.E., 2023. Senolytic therapy in mild Alzheimer's disease: a phase 1 feasibility trial. *Nat. Med* 29 (10), 2481–2488.
- González-Gualda, E., Páez-Ribes, M., Lozano-Torres, B., Macías, D., Wilson, J.R., González-López, C., Ou, H.-L., Mirón-Barroso, S., Zhang, Z., Lérica-Viso, A., Blandez, J.F., Bernardos, A., Sancenón, F., Rovira, M., Fruk, L., Martins, C.P., Serrano, M., Doherty, G.J., Martínez-Máñez, R., Muñoz-Espín, D., 2020. Galactose-conjugation of Navitoclax as an efficient strategy to increase senolytic specificity and reduce platelet toxicity. *Aging Cell* 19 (4), e13142.
- Görg, B., Karababa, A., Shafigullina, A., Bidmon, H.J., Häussinger, D., 2015. Ammonia-induced senescence in cultured rat astrocytes and in human cerebral cortex in hepatic encephalopathy. *Glia* 63 (1), 37–50.
- Gouw, A.A., van der Flier, W.M., Fazekas, F., van Straaten, E.C.W., Pantoni, L., Poggesi, A., Inzitari, D., Erkinjuntti, T., Wahlund, L.O., Waldemar, G., Schmidt, R., Scheltens, P., Barkhof, F., 2008. Progression of white matter hyperintensities and incidence of new lacunes over a 3-year period: the Leukoaraiosis and Disability study. *Stroke* 39 (5), 1414–1420.
- Graves, S.L., Meyer, C.F., Jeganathan, K.B., Baker, D.J., 2025. p16-expressing microglia and endothelial cells promote tauopathy and neurovascular abnormalities in PS19 mice. *Neuron*.
- Griveau, A., Wiel, C., Ziegler, D.V., Berge, M.O., Bernard, D., 2020. The JAK1/2 inhibitor ruxolitinib delays premature aging phenotypes. *Aging Cell* 19 (4), e13122.
- Grünberg, F., 2015. Invited review: drug development for tauopathies. *Neuropathol. Appl. Neurobiol.* 41 (1), 81–96.
- Guerrero, A., Herranz, N., Sun, B., Wagner, V., Gallage, S., Guiho, R., Wolter, K., Pombo, J., Irvine, E.E., Innes, A.J., Birch, J., Glegola, J., Manshaei, S., Heide, D., Dharmalingam, G., Harbig, J., Olona, A., Behmoaras, J., Dauch, D., Uren, A.G., Zender, L., Vernia, S., Martínez-Barbera, J.P., Heikenwalder, M., Withers, D.J., Gil, J., 2019. Cardiac glycosides are broad-spectrum senolytics. *Nat. Metab.* 1 (11), 1074–1088.
- Guerrero, A., Guiho, R., Herranz, N., Uren, A., Withers, D.J., Martínez-Barbera, J.P., Tietze, L.F., Gil, J., 2020. Galactose-modified duocarmycin prodrugs as senolytics. *Aging Cell* 19 (4), e13133.
- Guerrero, A., Innes, A.J., Roux, P.-F., Buisman, S.C., Jung, J., Ortet, L., Moiseeva, V., Wagner, V., Robinson, L., Ausema, A., Potapova, A., Perdiguero, E., Weersing, E., Aarts, M., Martin, N., Wuestefeld, T., Muñoz-Cánoves, P., de Haan, G., Bischof, O., Gil, J., 2022. 3-deazaadenosine (3dA) alleviates senescence to promote cellular fitness and cell therapy efficiency in mice. *Nat. Aging* 2, 851–866.
- Han, D.-W., Lee, M.H., Kim, B., Lee, J.J., Hyon, S.-H., Park, J.-C., 2012. Preventive effects of epigallocatechin-3-O-gallate against replicative senescence associated with p53 acetylation in human dermal fibroblasts. *Oxid. Med Cell Longev.* 2012, 850684.
- Han, L., Schuringa, J.J., Mulder, A., Vellenga, E., 2010. Dasatinib impairs long-term expansion of leukemic progenitors in a subset of acute myeloid leukemia cases. *Ann. Hematol.* 89 (9), 861–871.
- Han, X., Zhang, T., Liu, H., Mi, Y., Gou, X., 2020. Astrocyte senescence and alzheimer's disease: a review. *Front Aging Neurosci.* 12, 148.
- Hanan, E.J., Braun, M.-G., Heald, R.A., MacLeod, C., Chan, C., Clausen, S., Edgar, K.A., Eigenbrot, C., Elliott, R., Endres, N., Friedman, L.S., Gogol, E., Gu, X.-H., Thibodeau, R.H., Jackson, P.S., Kiefer, J.R., Knight, J.D., Nannini, M., Narukulla, R., Pace, A., Pang, J., Purkey, H.E., Salphati, L., Sampath, D., Schmidt, S., Sideris, S., Song, K., Sujatha-Bhaskar, S., Ultsch, M., Wallweber, H., Xin, J., Yeap, S., Young, A., Zhong, Y., Staben, S.T., 2022. Discovery of GDC-0077 (Inavolisib), a highly selective inhibitor and degrader of mutant PI3K α . *J. Med. Chem.* 65 (24), 16589–16621.
- Harley, C.B., Futcher, A.B., Greider, C.W., 1990. Telomeres shorten during ageing of human fibroblasts. *Nature* 345 (6274), 458–460.
- Harrison, C.N., Talpaz, M., Mead, A.J., 2016. Ruxolitinib is effective in patients with intermediate-1 risk myelofibrosis: a summary of recent evidence. *Leuk. Lymphoma* 57 (10), 2259–2267.
- Hartman, T.K., Wengenack, T.M., Poduslo, J.F., van Deursen, J.M., 2007. Mutant mice with small amounts of BubR1 display accelerated age-related gliosis. *Neurobiol. Aging* 28 (6), 921–927.
- Hayflick, L., 1965. The limited in vitro lifetime of human diploid cell strains. *Exp. Cell Res* 37, 614–636.
- He, N., Jin, W.L., Lok, K.H., Wang, Y., Yin, M., Wang, Z.J., 2013. Amyloid- β (1-42) oligomer accelerates senescence in adult hippocampal neural stem/progenitor cells via formylpeptide receptor 2. *Cell Death Dis.* 4 (11), e924.
- He, Y., Li, W., Lv, D., Zhang, X., Zhang, X., Ortiz, Y.T., Budamagunta, V., Campisi, J., Zheng, G., Zhou, D., 2020. Inhibition of USP7 activity selectively eliminates senescent cells in part via restoration of p53 activity. *Aging Cell* 19 (3), e13117.
- Hernandez-Guillamon, M., Mawhirt, S., Blais, S., Montaner, J., Neubert, T.A., Rostagno, A., Ghiso, J., 2015. Sequential amyloid- β degradation by the matrix metalloproteases MMP-2 and MMP-9. *J. Biol. Chem.* 290 (24), 15078–15091.
- Herranz, N., Gil, J., 2018. Mechanisms and functions of cellular senescence. *J. Clin. Invest* 128 (4), 1238–1246.
- Hou, J., Cui, C., Kim, S., Sung, C., Choi, C., 2018. Ginsenoside F1 suppresses astrocytic senescence-associated secretory phenotype. *Chem. Biol. Inter.* 283, 75–83.
- Hu, L., Li, H., Zi, M., Li, W., Liu, J., Yang, Y., Zhou, D., Kong, Q.-P., Zhang, Y., He, Y., 2022. Why senescent cells are resistant to apoptosis: an insight for senolytic development. *Front Cell Dev. Biol.* 10, 822816.
- Hubackova, S., Davidova, E., Rohlenova, K., Stursa, J., Werner, L., Andera, L., Dong, L., Terp, M.G., Hodny, Z., Ditzel, H.J., Rohlena, J., Neuzil, J., 2019. Selective elimination of senescent cells by mitochondrial targeting is regulated by ANT2. *Cell Death Differ.* 26 (2), 276–290.
- Hubbard, B.P., Sinclair, D.A., 2014. Small molecule SIRT1 activators for the treatment of aging and age-related diseases. *Trends Pharm. Sci.* 35 (3), 146–154.
- Jeon, O.H., Kim, C., Laberge, R.-M., Demaria, M., Rathod, S., Vasserot, A.P., Chung, J.W., Kim, D.H., Poon, Y., David, N., Baker, D.J., van Deursen, J.M., Campisi, J., Elisseeff, J.H., 2017. Local clearance of senescent cells attenuates the development of post-traumatic osteoarthritis and creates a pro-regenerative environment. *Nat. Med* 23 (6), 775–781.
- Jurk, D., Wang, C., Miwa, S., Maddick, M., Korolchuk, V., Tzolou, A., Gonos, E.S., Thrasivoulou, C., Saffrey, M.J., Cameron, K., von Zglinicki, T., 2012. Postmitotic neurons develop a p21-dependent senescence-like phenotype driven by a DNA damage response. *Aging Cell* 11 (6).

- Kang, C., Xu, Q., Martin, T.D., Li, M.Z., Demaria, M., Aron, L., Lu, T., Yankner, B.A., Campisi, J., Elledge, S.J., 2015. The DNA damage response induces inflammation and senescence by inhibiting autophagy of GATA4. *Science* 349 (6255), aaa5612.
- Kang, H.T., Park, J.T., Choi, K., Kim, Y., Choi, H.J.C., Jung, C.W., Lee, Y.-S., Park, S.C., 2017. Chemical screening identifies ATM as a target for alleviating senescence. *Nat. Chem. Biol.* 13 (6), 616–623.
- Khacho, M., Harris, R., Slack, R.S., 2019. Mitochondria as central regulators of neural stem cell fate and cognitive function. *Nat. Rev. Neurosci.* 20 (1), 34–48.
- Kim, E.N., Lim, J.H., Kim, M.Y., Ban, T.H., Jang, I.-A., Yoon, H.E., Park, C.W., Chang, Y. S., Choi, B.S., 2018a. Resveratrol, an Nrf2 activator, ameliorates aging-related progressive renal injury. *Aging (Albany NY)* 10 (1), 83–99.
- Kim, N., Do, J., Bae, J.-S., Jin, H.K., Kim, J.-H., Inn, K.-S., Oh, M.S., Lee, J.K., 2018b. Piperlongumine inhibits neuroinflammation via regulating NF- κ B signaling pathways in lipopolysaccharide-stimulated BV2 microglia cells. *J. Pharm. Sci.* 137 (2), 195–201.
- Kim, T.H., Song, J., Kim, S.-H., Parikh, A.K., Mo, X., Palanichamy, K., Kaur, B., Yu, J., Yoon, S.O., Nakano, I., Kwon, C.-H., 2014. Piperlongumine treatment inactivates peroxiredoxin 4, exacerbates endoplasmic reticulum stress, and preferentially kills high-grade glioma cells. *Neuro Oncol.* 16 (10), 1354–1364.
- Kiryk, A., Sowodniok, K., Kreiner, G., Rodriguez-Parkitna, J., Sönmez, A., Górkiewicz, T., Bierhoff, H., Wawrzyniak, M., Janusz, A.K., Liss, B., Konopka, W., Schütz, G., Kaczmarek, L., Parlato, R., 2013. Impaired rRNA synthesis triggers homeostatic responses in hippocampal neurons. *Front. Cell Neurosci.* 7, 207.
- Kiss, T., Nyúl-Tóth, Á., Balasubramanian, P., Tarantini, S., Ahire, C., Yabluchanskiy, A., Csipo, T., Farkas, E., Wren, J.D., Garman, L., Csiszar, A., Ungvari, Z., 2020. Nicotinamide mononucleotide (NMN) supplementation promotes neurovascular rejuvenation in aged mice: transcriptional footprint of SIRT1 activation, mitochondrial protection, anti-inflammatory, and anti-apoptotic effects. *Geroscience* 42 (2), 527–546.
- Kiyono, T., Foster, S.A., Koop, J.I., McDougall, J.K., Galloway, D.A., Klingelutz, A.J., 1998. Both Rb/p16INK4a inactivation and telomerase activity are required to immortalize human epithelial cells. *Nature* 396 (6706), 84–88.
- Knopp, R.C., Erickson, M.A., Rhea, E.M., Reed, M.J., Banks, W.A., 2023. Cellular senescence and the blood-brain barrier: Implications for aging and age-related diseases. *Exp. Biol. Med. (Maywood)* 248 (5), 399–411.
- Kolacz, K., Robaszekiewicz, A., 2024. PARP1 at the crossroad of cellular senescence and nucleolar processes. *Ageing Res. Rev.* 94, 102206.
- Kujiro, Y., Suzuki, N., Kondo, T., 2010. Esophageal cancer-related gene 4 is a secreted inducer of cell senescence expressed by aged CNS precursor cells. *Proc. Natl. Acad. Sci. USA* 107 (18), 8259–8264.
- Kuk, M.U., Kim, J.W., Lee, Y.-S., Cho, K.A., Park, J.T., Park, S.C., 2019. Alleviation of senescence via ATM inhibition in accelerated aging models. *Mol. Cells* 42 (3), 210–217.
- Kumar, R., Sharma, A., Kumari, A., Gulati, A., Padwad, Y., Sharma, R., 2019. Epigallocatechin gallate suppresses premature senescence of preadipocytes by inhibition of PI3K/Akt/mTOR pathway and induces senescent cell death by regulation of Bax/Bcl-2 pathway. *Biogerontology* 20 (2), 171–189.
- Labege, R.-M., Zhou, L., Sarantos, M.R., Rodier, F., Freund, A., de Keizer, P.L.J., Liu, S., Demaria, M., Cong, Y.-S., Kapahi, P., Desprez, P.-Y., Hughes, R.E., Campisi, J., 2012. Glucocorticoids suppress selected components of the senescence-associated secretory phenotype. *Aging Cell* 11 (4), 569–578.
- Labege, R.-M., Sun, Y., Orjalo, A.V., Patil, C.K., Freund, A., Zhou, L., Curran, S.C., Dávalos, A.R., Wilson-Edell, K.A., Liu, S., Limbad, C., Demaria, M., Li, P., Hubbard, G.B., Ikono, Y., Javors, M., Desprez, P.-Y., Benz, C.C., Kapahi, P., Nelson, P. S., Campisi, J., 2015. MTOR regulates the pro-tumorigenic senescence-associated secretory phenotype by promoting IL1A translation. *Nat. Cell Biol.* 17 (8), 1049–1061.
- de Lange, T., 1998. Telomeres and senescence: ending the debate. *Science* 279 (5349), 334–335.
- Lau, V., Ramer, L., Tremblay, M.-È., 2023. An aging, pathology burden, and glial senescence build-up hypothesis for late onset Alzheimer's disease. *Nat. Commun.* 14 (1), 1670.
- Lee, J.-H., Yu, W.H., Kumar, A., Lee, S., Mohan, P.S., Peterhoff, C.M., Wolfe, D.M., Martinez-Vicente, M., Massey, A.C., Sovak, G., Uchiyama, Y., Westaway, D., Cuervo, A.M., Nixon, R.A., 2010. Lysosomal proteolysis and autophagy require presenilin 1 and are disrupted by Alzheimer-related PS1 mutations. *Cell* 141 (7), 1146–1158.
- Lee, Y.R., Cho, H.M., Park, E.J., Zhang, M., Doan, T.P., Lee, B.W., Cho, K.A., Oh, W.K., 2020. Metabolite profiling of rambutan (*Nephelium lappaceum* L.) seeds using UPLC-qTOF-MS/MS and senomorphic effects in aged human dermal fibroblasts. *Nutrients* 12 (5).
- Lérida-Viso, A., Estepa-Fernández, A., Morellá-Aucejo, Á., Lozano-Torres, B., Alfonso, M., Blandez, J.F., Bisbal, V., Sepúlveda, P., García-Fernández, A., Orzáez, M., Martínez-Máñez, R., 2022. Pharmacological senolysis reduces doxorubicin-induced cardiotoxicity and improves cardiac function in mice. *Pharm. Res.* 183, 106356.
- Levenson, J.D., Phillips, D.C., Mitten, M.J., Boghaert, E.R., Diaz, D., Tahir, S.K., Belmont, L.D., Nimmer, P., Xiao, Y., Ma, X.M., Lowes, K.N., Kovar, P., Chen, J., Jin, S., Smith, M., Xue, J., Zhang, H., Oleksijew, A., Magoc, T.J., Vaidya, K.S., Albert, D.H., Tarrant, J.M., La, N., Wang, L., Tao, Z.-F., Wendt, M.D., Sampath, D., Rosenberg, S.H., Tse, C., Huang, D.C.S., Fairbrother, W.J., Elmore, S.W., Souers, A.J., 2015. Exploiting selective BCL-2 family inhibitors to dissect cell survival dependencies and define improved strategies for cancer therapy. *Sci. Transl. Med.* 7 (279), 279ra240.
- Lewis, L., 2022. Saf. Feasibility Dasatinib Quercetin Adults Risk Alzheimer's Dis. (STAMINA). (<https://clinicaltrials.gov/study/NCT05422885>study-record-dates).
- Li, J., Rix, U., Fang, B., Bai, Y., Edwards, A., Colinge, J., Bennett, K.L., Gao, J., Song, L., Eschrich, S., Superti-Furga, G., Koomen, J., Haura, E.B., 2010. A chemical and phosphoproteomic characterization of dasatinib action in lung cancer. *Nat. Chem. Biol.* 6 (4), 291–299.
- Li, S., Sun, M., Hao, C., Qu, A., Wu, X., Xu, L., Xu, C., Kuang, H., 2020. Chiral Cux Coy S nanoparticles under magnetic field and NIR light to eliminate senescent cells. *Angew. Chem. Int. Ed. Engl.* 59 (33), 13915–13922.
- Li, W., He, Y., Zhang, R., Zheng, G., Zhou, D., 2019. The curcumin analog EF24 is a novel senolytic agent. *Aging (Albany NY)* 11 (2), 771–782.
- Li, W.-Q., Wang, Z.-J., Liu, S., Hu, Y., Yin, M., Lu, Y., 2016. N-stearoyl-L-tyrosine inhibits the senescence of neural stem/progenitor cells induced by A β 1–42 via the CB2 receptor. *Stem Cells Int* 2016, 7419389.
- Liu, G.-H., Qu, J., Shen, X., 2008. NF-kappaB/p65 antagonizes Nrf2-ARE pathway by depriving CBP from Nrf2 and facilitating recruitment of HDAC3 to MafK. *Biochim Biophys. Acta* 1783 (5), 713–727.
- Liu, X., Wang, Y., Zhang, X., Gao, Z., Zhang, S., Shi, P., Zhang, X., Song, L., Hendrickson, H., Zhou, D., Zheng, G., 2018. Senolytic activity of piperlongumine analogues: Synthesis and biological evaluation. *Bioorg. Med. Chem.* 26 (14), 3925–3938.
- Long, S., Benoist, C., Weidner, W., 2023. World Alzheimer Report 2023: reducing dementia risk: never too early, never too late. Alzheimer's Disease International, London, England.
- Ma, X.-R., Zhu, X., Xiao, Y., Gu, H.-M., Zheng, S.-S., Li, L., Wang, F., Dong, Z.-J., Wang, D.-X., Wu, Y., Yang, C., Jiang, W., Yao, K., Yin, Y., Zhang, Y., Peng, C., Gao, L., Meng, Z., Hu, Z., Liu, C., Li, L., Chen, H.-Z., Shu, Y., Ju, Z., Zhao, J.-W., 2022. Restoring nuclear entry of Sirtuin 2 in oligodendrocyte progenitor cells promotes remyelination during ageing. *Nat. Commun.* 13 (1), 1225.
- Maina, M.B., Bailey, L.J., Wagih, S., Biasetti, L., Pollack, S.J., Quinn, J.P., Thorpe, J.R., Doherty, A.J., Serpell, L.C., 2018. The involvement of tau in nucleolar transcription and the stress response. *Acta Neuropathol. Commun.* 6 (1), 70.
- Malaquin, N., Martinez, A., Rodier, F., 2016. Keeping the senescence secretome under control: molecular reins on the senescence-associated secretory phenotype. *Exp. Gerontol.* 82, 39–49.
- Mao, B., Zhang, Q., Ma, L., Zhao, D.-S., Zhao, P., Yan, P., 2022. Overview of research into mTOR inhibitors. *Molecules* 27 (16).
- Martínez-Zamudio, R.I., Robinson, L., Roux, P.-F., Bischof, O., 2017. Snapshot: cellular senescence pathways. *Cell* 170 (4).
- Matsumoto, T., Baker, D.J., d'Uscio, L.V., Mozammel, G., Katusic, Z.S., van Deursen, J.M., 2007. Aging-associated vascular phenotype in mutant mice with low levels of BubR1. *Stroke* 38 (3), 1050–1056.
- Menicacci, B., Cipriani, C., Margheri, F., Mocali, A., Giovannelli, L., 2017. Modulation of the senescence-associated inflammatory phenotype in human fibroblasts by olive phenols. *Int. J. Mol. Sci.* 18 (11).
- Mérino, D., Khaw, S.L., Glaser, S.P., Anderson, D.J., Belmont, L.D., Wong, C., Yue, P., Robati, M., Phipson, B., Fairlie, W.D., Lee, E.F., Campbell, K.J., Vandenberg, C.J., Cory, S., Roberts, A.W., Ludlam, M.J.C., Huang, D.C.S., Bouillet, P., 2012. Bcl-2, Bcl-x(L), and Bcl-w are not equivalent targets of ABT-737 and navitoclax (ABT-263) in lymphoid and leukemic cells. *Blood* 119 (24), 5807–5816.
- Millar, C.L., Iloputaife, I., Baldyga, K., Norling, A.M., Boulougoura, A., Vichos, T., Tchkonina, T., Deisinger, A., Pirtskhalava, T., Kirkland, J.L., Trivison, T.G., Lipsitz, L. A., 2025. A pilot study of senolytics to improve cognition and mobility in older adults at risk for Alzheimer's disease. *EBioMedicine* 113, 105612.
- Minor, R.K., Baur, J.A., Gomes, A.P., Ward, T.M., Csiszar, A., Mercken, E.M., Abdelmohsen, K., Shin, Y.-K., Canto, C., Scheibye-Knudsen, M., Krawczyk, M., Irusta, P.M., Martin-Montalvo, A., Hubbard, B.P., Zhang, Y., Lehmann, E., White, A. A., Price, N.L., Swindell, W.R., Pearson, K.J., Becker, K.G., Bohr, V.A., Gorospe, M., Egan, J.M., Talan, M.L., Auwerx, J., Westphal, C.H., Ellis, J.L., Ungvari, Z., Vlasuk, G. P., Elliott, P.J., Sinclair, D.A., de Cabo, R., 2011. SRT1720 improves survival and healthspan of obese mice. *Sci. Rep.* 1, 70.
- Mitchell, S.J., Martin-Montalvo, A., Mercken, E.M., Palacios, H.H., Ward, T.M., Abulwerdi, G., Minor, R.K., Vlasuk, G.P., Ellis, J.L., Sinclair, D.A., Dawson, J., Allison, D.B., Zhang, Y., Becker, K.G., Bernier, M., de Cabo, R., 2014. The SIRT1 activator SRT1720 extends lifespan and improves health of mice fed a standard diet. *Cell Rep.* 6 (5), 836–843.
- Moiseeva, O., Deschênes-Simard, X., St-Germain, E., Igelmann, S., Huot, G., Cadar, A.E., Bourdeau, V., Pollak, M.N., Ferbeyre, G., 2013. Metformin inhibits the senescence-associated secretory phenotype by interfering with IKK/NF- κ B activation. *Aging Cell* 12 (3), 489–498.
- Mombach, J.C.M., Vendrusculo, B., Bugs, C.A., 2015. A model for p38MAPK-induced astrocyte senescence. *PLoS One* 10 (5), e0125217.
- Moncsek, A., Al-Suraihi, M.S., Trusconi, C.E., O'Hara, S.P., Splinter, P.L., Zuber, C., Patsenker, E., Valli, P.V., Fingas, C.D., Weber, A., Zhu, Y., Tchkonina, T., Kirkland, J. L., Gores, G.J., Müllhaupt, B., LaRusso, N.F., Mertens, J.C., 2018. Targeting senescent cholangiocytes and activated fibroblasts with B-cell lymphoma-extra large inhibitors ameliorates fibrosis in multidrug resistance 2 gene knockout (Mdr2^{-/-}) mice. *Hepatology* 67 (1), 247–259.
- Montagne, A., Zhao, Z., Zlokovic, B.V., 2017. Alzheimer's disease: a matter of blood-brain barrier dysfunction? *J. Exp. Med.* 214 (11), 3151–3169.
- Montero, J.C., Seoane, S., Ocaña, A., Pandiella, A., 2011. Inhibition of SRC family kinases and receptor tyrosine kinases by dasatinib: possible combinations in solid tumors. *Clin. Cancer Res.* 17 (17), 5546–5552.
- Moreno-García, A., Kun, A., Calero, O., Medina, M., Calero, M., 2018. An overview of the role of lipofuscin in age-related neurodegeneration. *Front. Neurosci.* 12, 464.
- Mummary, C.J., Börjesson-Hanson, A., Blackburn, D.J., Vijverberg, E.G.B., De Deyn, P.P., Ducharme, S., Jonsson, M., Schneider, A., Rinne, J.O., Ludolph, A.C., Bodenschatz, R., Kordasiewicz, H., Swayze, E.E., Fitzsimmons, B., Mignon, L.,

- Moore, K.M., Yun, C., Baumann, T., Li, D., Norris, D.A., Crean, R., Graham, D.L., Huang, E., Ratti, E., Bennett, C.F., Junge, C., Lane, R.M., 2023. Tau-targeting antisense oligonucleotide MAPTRx in mild Alzheimer's disease: a phase 1b, randomized, placebo-controlled trial. *Nat. Med.* 29 (6), 1437–1447.
- Musi, N., Valentine, J.M., Sickora, K.R., Baeuerle, E., Thompson, C.S., Shen, Q., Orr, M. E., 2018. Tau protein aggregation is associated with cellular senescence in the brain. *Aging Cell* 17 (6), e12840.
- Nadile, M., Retsidou, M.I., Gioti, K., Beloukas, A., Tsiani, E., 2022. Resveratrol against cervical cancer: evidence from in vitro and in vivo studies. *Nutrients* 14 (24).
- Nakamura, A.J., Chiang, Y.J., Hathcock, K.S., Horikawa, I., Sedelnikova, O.A., Hodes, R. J., Bonner, W.M., 2008. Both telomeric and non-telomeric DNA damage are determinants of mammalian cellular senescence. *Epigenetics Chromatin* 1 (1), 6.
- Nambiar, A., Kellogg, D., Justice, J., Goros, M., Gelfond, J., Pascual, R., Hashmi, S., Masternak, M., Prata, L., LeBrasseur, N., Limper, A., Kritchevsky, S., Musi, N., Tchkonina, T., Kirkland, J., 2023. Senolytics dasatinib and quercetin in idiopathic pulmonary fibrosis: results of a phase I, single-blind, single-center, randomized, placebo-controlled pilot trial on feasibility and tolerability. *EBioMedicine* 90, 104481.
- Nasrabady, S.E., Rizvi, B., Goldman, J.E., Brickman, A.M., 2018. White matter changes in Alzheimer's disease: a focus on myelin and oligodendrocytes. *Acta Neuropathol. Commun.* 6 (1), 22.
- Nation, D.A., Sweeney, M.D., Montagne, A., Sagare, A.P., D'Orazio, L.M., Pachicano, M., Sepeshband, F., Nelson, A.R., Buennagel, D.P., Harrington, M.G., Benzinger, T.L.S., Fagan, A.W., Ringman, J.M., Schneider, L.S., Morris, J.C., Chui, H.C., Law, M., Toga, A.M., Zlokovic, B.V., 2019. Blood-brain barrier breakdown is an early biomarker of human cognitive dysfunction. *Nat. Med.* 25 (2), 270–276.
- Nicaise, A.M., Wagstaff, L.J., Willis, C.M., Paisie, C., Chandok, H., Robson, P., Fossati, V., Williams, A., Crocker, S.J., 2019. Cellular senescence in progenitor cells contributes to diminished remyelination potential in progressive multiple sclerosis. *Proc. Natl. Acad. Sci. USA* 116 (18), 9030–9039.
- Nie, X., Liang, L., Xi, H., Jiang, S., Jiang, J., Tang, C., Liu, X., Liu, S., Wan, C., Zhao, J., Yang, J., 2015. 2, 3, 7, 8-Tetrachlorodibenzo-p-dioxin induces premature senescence of astrocytes via WNT/ β -catenin signaling and ROS production. *J. Appl. Toxicol.* 35 (7), 851–860.
- Niu, Q., Li, D., Zhang, J., Piao, Z., Xu, B., Xi, Y., Mohamed Kamal, N.N.S.N., Lim, V., Li, P., Yin, Y., 2024. The new perspective of Alzheimer's disease research: mechanism and therapeutic strategy of neuronal senescence. *Ageing Res. Rev.* 102, 102593.
- Ogrodnik, M., Miwa, S., Tchkonina, T., Tiniakos, D., Wilson, C.L., Lahat, A., Day, C.P., Burt, A., Palmer, A., Anstee, Q.M., Grellescheid, S.N., Hoeijmakers, J.H.J., Barnhoorn, S., Mann, D.A., Bird, T.G., Vermeij, W.P., Kirkland, J.L., Passos, J.F., von Zglinicki, T., Jurk, D., 2017. Cellular senescence drives age-dependent hepatic steatosis. *Nat. Commun.* 8, 15691.
- Ogrodnik, M., Zhu, Y., Langhi, L.G.P., Tchkonina, T., Krüger, P., Fielder, E., Victorelli, S., Ruswhandi, R.A., Giorgadze, N., Pirtskhalava, T., Podgorni, O., Enikolopov, G., Johnson, K.O., Xu, M., Inman, C., Palmer, A.K., Schafer, M., Weigl, M., Ikeno, Y., Burns, T.C., Passos, J.F., von Zglinicki, T., Kirkland, J.L., Jurk, D., 2019b. Obesity-induced cellular senescence drives anxiety and impairs neurogenesis. *Cell Metab.* 29 (5).
- Ogrodnik, M., Zhu, Y., Langhi, L.G.P., Tchkonina, T., Krüger, P., Fielder, E., Victorelli, S., Ruswhandi, R.A., Giorgadze, N., Pirtskhalava, T., Podgorni, O., Enikolopov, G., Johnson, K.O., Xu, M., Inman, C., Palmer, A.K., Schafer, M., Weigl, M., Ikeno, Y., Burns, T.C., Passos, J.F., von Zglinicki, T., Kirkland, J.L., Jurk, D., 2019a. Obesity-induced cellular senescence drives anxiety and impairs neurogenesis. *Cell Metab.* 29 (5), 1233.
- Ohashi, M., Korsakova, E., Allen, D., Lee, P., Fu, K., Vargas, B.S., Cinkornpumin, J., Salas, C., Park, J.C., Germanguz, I., Langerman, J., Chronis, C., Kuoy, E., Tran, S., Xiao, X., Pellegrini, M., Plath, K., Lowry, W.E., 2018. Loss of MEC2 leads to activation of P53 and neuronal senescence. *Stem Cell Rep.* 10 (5), 1453–1463.
- Oltersdorf, T., Elmore, S.W., Shoemaker, A.R., Armstrong, R.C., Augeri, D.J., Belli, B.A., Bruncko, M., Deckwerth, T.L., Dinges, J., Hajduk, P.J., Joseph, M.K., Kitada, S., Korsmeyer, S.J., Kunzer, A.R., Letai, A., Li, C., Mitten, M.J., Nettesheim, D.G., Ng, S., Nimmer, P.M., O'Connor, J.M., Oleksijew, A., Petros, A.M., Reed, J.C., Shen, W., Tahir, S.K., Thompson, C.B., Tomaselli, K.J., Wang, B., Wendt, M.D., Zhang, H., Fesik, S.W., Rosenberg, S.H., 2005. An inhibitor of Bcl-2 family proteins induces regression of solid tumours. *Nature* 435 (7042), 677–681.
- Ota, H., Akishita, M., Akiyoshi, T., Kahyo, T., Setou, M., Ogawa, S., Iijima, K., Eto, M., Ouchi, Y., 2012. Testosterone deficiency accelerates neuronal and vascular aging of SAMP8 mice: protective role of eNOS and SIRT1. *PLoS One* 7 (1), e29598.
- Owen, M.R., Doran, E., Halestrap, A.P., 2000. Evidence that metformin exerts its anti-diabetic effects through inhibition of complex I of the mitochondrial respiratory chain. *Biochem. J.* 348 (Pt 3 (Pt 3)), 607–614.
- Pan, J., Li, D., Xu, Y., Zhang, J., Wang, Y., Chen, M., Lin, S., Huang, L., Chung, E.J., Citrin, D.E., Wang, Y., Hauer-Jensen, M., Zhou, D., Meng, A., 2017. Inhibition of Bcl-2/xl with ABT-263 selectively kills senescent type II pneumocytes and reverses persistent pulmonary fibrosis induced by ionizing radiation in mice. *Int. J. Radiat. Oncol. Biol. Phys.* 99 (2), 353–361.
- Panosian, L., Fenik, P., Zhu, Y., Zhan, G., McBurney, M.W., Veasey, S., 2011. SIRT1 regulation of wakefulness and senescence-like phenotype in wake neurons. *J. Neurosci.* 31 (11), 4025–4036.
- Papadopol, D., Boulay, K., Kazak, L., Pollak, M., Mallette, F.A., Topisirovic, I., Hulea, L., 2019. mTOR as a central regulator of lifespan and aging. *F1000Res* 8.
- Perrott, K.M., Wiley, C.D., Desprez, P.-Y., Campisi, J., 2017. Apigenin suppresses the senescence-associated secretory phenotype and paracrine effects on breast cancer cells. *Geroscience* 39 (2), 161–173.
- Pertusa, M., García-Matas, S., Rodríguez-Farré, E., Sanfeliu, C., Cristófol, R., 2007. Astrocytes aged in vitro show a decreased neuroprotective capacity. *J. Neurochem.* 101 (3), 794–805.
- Pitozzi, V., Mocali, A., Laurenzana, A., Giannoni, E., Cifola, I., Battaglia, C., Chiarugi, P., Dolara, P., Giovannelli, L., 2013. Chronic resveratrol treatment ameliorates cell adhesion and mitigates the inflammatory phenotype in senescent human fibroblasts. *J. Gerontol. A Biol. Sci. Med. Sci.* 68 (4), 371–381.
- Preininger, M.K., Zaytseva, D., Lin, J.M., Kaufer, D., 2023. Blood-brain barrier dysfunction promotes astrocyte senescence through albumin-induced TGF β signaling activation. *Aging Cell* 22 (2), e13747.
- Quan, S., Fu, X., Cai, H., Ren, Z., Xu, Y., Jia, L., 2025. The neuroimmune nexus: unraveling the role of the mtDNA-cGAS-STING signal pathway in Alzheimer's disease. *Mol. Neurodegener.* 20 (1), 25.
- Rabinovici, G.D., 2021. Controversy and progress in Alzheimer's disease - FDA approval of aducanumab. *N. Engl. J. Med.* 385 (9), 771–774.
- Rachmian, N., Medina, S., Cherqui, U., Akiva, H., Deitch, D., Edilbi, D., Croese, T., Salame, T.M., Ramos, J.M.P., Cahalon, L., Krizhanovsky, V., Schwartz, M., 2024. Identification of senescent, TREM2-expressing microglia in aging and Alzheimer's disease model mouse brain. *Nat. Neurosci.* 27 (6), 1116–1124.
- Ransohoff, R.M., Brown, M.A., 2012. Innate immunity in the central nervous system. *J. Clin. Invest.* 122 (4), 1164–1171.
- Reynolds, R., Dawson, M., Papadopoulos, D., Polito, A., Di Bello, I.C., Pham-Dinh, D., Levine, J., 2002. The response of NG2-expressing oligodendrocyte progenitors to demyelination in MOG-EAE and MS. *J. Neurocytol.* 31 (6-7), 523–536.
- Richardson, B., Goedert, T., Quraishie, S., Deinhardt, K., Mudher, A., 2023. How do neurons age? A focused review on the aging of the microtubular cytoskeleton. *Neural Regen. Res* 19 (9), 1899–1907.
- Rim, C., You, M.-J., Nahm, M., Kwon, M.-S., 2024. Emerging role of senescent microglia in brain aging-related neurodegenerative diseases. *Transl. Neurodegener.* 13 (1), 10.
- Rochford, K.D., Collins, L.E., McLoughlin, A., Cummins, P.M., 2015. Tumour necrosis factor- α -mediated disruption of cerebrovascular endothelial barrier integrity in vitro involves the production of proinflammatory interleukin-6. *J. Neurochem.* 136 (3), 564–572.
- Rodier, F., Campisi, J., 2011. Four faces of cellular senescence. *J. Cell Biol.* 192 (4), 547–556.
- Rodier, F., Coppé, J.-P., Patil, C.K., Hoeijmakers, W.A.M., Muñoz, D.P., Raza, S.R., Freund, A., Campeau, E., Davalos, A.R., Campisi, J., 2009. Persistent DNA damage signalling triggers senescence-associated inflammatory cytokine secretion. *Nat. Cell Biol.* 11 (8), 973–979.
- Roe, S.M., Prodromou, C., O'Brien, R., Ladbury, J.E., Piper, P.W., Pearl, L.H., 1999. Structural basis for inhibition of the Hsp90 molecular chaperone by the antitumor antibiotics radicicol and geldanamycin. *J. Med. Chem.* 42 (2), 260–266.
- Saccon, T.D., Nagpal, R., Yadav, H., Cavalcante, M.B., Nunes, A.D.C., Schneider, A., Gesing, A., Hughes, B., Yousefzadeh, M., Tchkonina, T., Kirkland, J.L., Niedernhofer, L.J., Robbins, P.D., Masternak, M.M., 2021. Senolytic combination of dasatinib and quercetin alleviates intestinal senescence and inflammation and modulates the gut microbiome in aged mice. *J. Gerontol. A Biol. Sci. Med. Sci.* 76 (11), 1895–1905.
- Saez-Atienzar, S., Masliah, E., 2020. Cellular senescence and Alzheimer disease: the egg and the chicken scenario. *Nat. Rev. Neurosci.* 21 (8), 433–444.
- Salminen, A., Ojala, J., Kaarniranta, K., Haapasalo, A., Hiltunen, M., Soininen, H., 2011. Astrocytes in the aging brain express characteristics of senescence-associated secretory phenotype. *Eur. J. Neurosci.* 34 (1).
- Samaraweera, L., Adomako, A., Rodriguez-Gabin, A., McDavid, H.M., 2017. A Novel indication for panobinostat as a senolytic drug in NSCLC and HNSCC. *Sci. Rep.* 7 (1), 1900.
- Saul, D., Jurk, D., Doolittle, M.L., Kosinsky, R.L., Han, Y., Zhang, X., Franco, A.C., Kim, S. Y., Wyles, S.P., Prakash, Y.S., Monroe, D.G., Ferrucci, L., LeBrasseur, N.K., Robbins, P.D., Niedernhofer, L.J., Khosla, S., Passos, J.F., 2025. Distinct senotypes in p16- and p21-positive cells across human and mouse aging tissues. *EMBO J.*
- Schafer, M.J., White, T.A., Iijima, K., Haak, A.J., Ligresti, G., Atkinson, E.J., Oberg, A.L., Birch, J., Salmonowicz, H., Zhu, Y., Mazula, D.L., Brooks, R.W., Fuhrmann-Stroissnigg, H., Pirtskhalava, T., Prakash, Y.S., Tchkonina, T., Robbins, P.D., Aubry, M.C., Passos, J.F., Kirkland, J.L., Tschumperlin, D.J., Kita, H., LeBrasseur, N. K., 2017. Cellular senescence mediates fibrotic pulmonary disease. *Nat. Commun.* 8, 14532.
- Schoenwaelder, S.M., Jarman, K.E., Gardiner, E.E., Hua, M., Qiao, J., White, M.J., Josefsson, E.C., Alwis, I., Ono, A., Willcox, A., Andrews, R.K., Mason, K.D., Salem, H. H., Huang, D.C.S., Kile, B.T., Roberts, A.W., Jackson, S.P., 2011. Bcl-xL-inhibitory BH3 mimetics can induce a transient thrombocytopenia that undermines the hemostatic function of platelets. *Blood* 118 (6), 1663–1674.
- Sedelnikova, O.A., Horikawa, I., Zimonjic, D.B., Popescu, N.C., Bonner, W.M., Barrett, J. C., 2004. Senescing human cells and ageing mice accumulate DNA lesions with unreparable double-strand breaks. *Nat. Cell Biol.* 6 (2), 168–170.
- Shanbhag, N.M., Evans, M.D., Mao, W., Nana, A.L., Seely, W.W., Adame, A., Rissman, R. A., Masliah, E., Mucke, L., 2019. Early neuronal accumulation of DNA double strand breaks in Alzheimer's disease. *Acta Neuropathol. Commun.* 7 (1), 77.
- Si, Z., Sun, L., Wang, X., 2021. Evidence and perspectives of cell senescence in neurodegenerative diseases. *Biomed. Pharm.* 137, 111327.
- Sierra, A., Gottfried-Blackmore, A.C., McEwen, B.S., Bulloch, K., 2007. Microglia derived from aging mice exhibit an altered inflammatory profile. *Glia* 55 (4), 412–424.
- Sims, J.R., Zimmer, J.A., Evans, C.D., Lu, M., Ardayfio, P., Sparks, J., Wessels, A.M., Shcherbinin, S., Wang, H., Monkul Nery, E.S., Collins, E.C., Solomon, P., Salloway, S., Apostolova, L.G., Hansson, O., Ritchie, C., Brooks, D.A., Mintun, M., Skovronsky, D.M., 2023. Donanemab in early symptomatic alzheimer disease: the TRAILBLAZER-ALZ 2 randomized clinical trial. *JAMA* 330 (6), 512–527.

- Smits, V.A.J., Reaper, P.M., Jackson, S.P., 2006. Rapid PIKK-dependent release of Chk1 from chromatin promotes the DNA-damage checkpoint response. *Curr. Biol.* 16 (2), 150–159.
- Streit, W.J., Braak, H., Xue, Q.-S., Bechmann, I., 2009. Dystrophic (senescent) rather than activated microglial cells are associated with tau pathology and likely precede neurodegeneration in Alzheimer's disease. *Acta Neuropathol.* 118 (4), 475–485.
- Suberbielle, E., Sanchez, P.E., Kravitz, A.V., Wang, X., Ho, K., Eilertson, K., Devidze, N., Kreitzer, A.C., Mucke, L., 2013. Physiologic brain activity causes DNA double-strand breaks in neurons, with exacerbation by amyloid- β . *Nat. Neurosci.* 16 (5), 613–621.
- Sun, S., Qin, W., Tang, X., Meng, Y., Hu, W., Zhang, S., Qian, M., Liu, Z., Cao, X., Pang, Q., Zhao, B., Wang, Z., Zhou, Z., Liu, B., 2020. Vascular endothelium-targeted Sirt7 gene therapy rejuvenates blood vessels and extends life span in a Hutchinson-Gilford progeria model. *Sci. Adv.* 6 (8), eay5556.
- Sun, Y., Wei, J., Ma, S., He, C., Sui, L., Pan, X., Zhu, X., 2025. Lysosomes as dynamic regulators of metabolic signaling and organ physiology in aging: from mechanism to therapy. *Aging Dis.*
- Suryadevara, V., Hudgins, A.D., Rajesh, A., Pappalardo, A., Karpova, A., Dey, A.K., Hertz, A., Agudelo, A., Rocha, A., Soygur, B., Schilling, B., Carver, C.M., Aguayo-Mazzucato, C., Baker, D.J., Bernlohr, D.A., Jurk, D., Mangarova, D.B., Quardokus, E. M., Enninga, E.A.L., Schmidt, E.L., Chen, F., Duncan, F.E., Cambuli, F., Kaur, G., Kuchel, G.A., Lee, G., Daldrop-Link, H.E., Martini, H., Phatnani, H., Al-Naggar, I.M., Rahman, I., Nie, J., Passos, J.F., Silverstein, J.C., Campisi, J., Wang, J., Iwasaki, K., Barbosa, K., Metis, K., Nernekli, K., Niedernhofer, L.J., Ding, L., Wang, L., Adams, L. C., Ruiyang, L., Doolittle, M.L., Teneche, M.G., Schafer, M.J., Xu, M., Hajipour, M., Boroumand, M., Basisty, N., Sloan, N., Slavov, N., Kuksenok, O., Robson, P., Gomez, P.T., Vasilikos, P., Adams, P.D., Carapeto, P., Zhu, Q., Ramasamy, R., Perez-Lorenzo, R., Fan, R., Dong, R., Montgomery, R.R., Shaikh, S., Vickovic, S., Yin, S., Kang, S., Suvakov, S., Khosla, S., Garovic, V.D., Menon, V., Xu, Y., Song, Y., Suh, Y., Dou, Z., Neretti, N., 2024. SenNet recommendations for detecting senescent cells in different tissues. *Nat. Rev. Mol. Cell Biol.* 25 (12), 1001–1023.
- Swanson, E.C., Manning, B., Zhang, H., Lawrence, J.B., 2013. Higher-order unfolding of satellite heterochromatin is a consistent and early event in cell senescence. *J. Cell Biol.* 203 (6), 929–942.
- Swanson, E.C., Rapkin, L.M., Bazett-Jones, D.P., Lawrence, J.B., 2015. Unfolding the story of chromatin organization in senescent cells. *Nucleus* 6 (4), 254–260.
- Sweeney, M.D., Sagare, A.P., Zlokovic, B.V., 2018. Blood-brain barrier breakdown in Alzheimer disease and other neurodegenerative disorders. *Nat. Rev. Neurol.* 14 (3), 133–150.
- Syed, D.N., Adhami, V.M., Khan, N., Khan, M.I., Mukhtar, H., 2016. Exploring the molecular targets of dietary flavonoid fisetin in cancer. *Semin Cancer Biol.* 40–41, 130–140.
- Tan, J.X., Finkel, T., 2023. Lysosomes in senescence and aging. *EMBO Rep.* 24 (11), e57265.
- Tavana, O., Benjamin, C.L., Puebla-Osorio, N., Sang, M., Ullrich, S.E., Ananthaswamy, H. N., Zhu, C., 2010. Absence of p53-dependent apoptosis leads to UV radiation hypersensitivity, enhanced immunosuppression and cellular senescence. *Cell Cycle* 9 (16), 3328–3336.
- Teo, Y.V., Rattanavirotkul, N., Olova, N., Salzano, A., Quintanilla, A., Tarrats, N., Kiourtsi, C., Müller, M., Green, A.R., Adams, P.D., Acosta, J.-C., Bird, T.G., Kirschner, K., Neretti, N., Chandra, T., 2019. Notch signaling mediates secondary senescence. *Cell Rep.* 27 (4).
- Terrando, N., Monaco, C., Ma, D., Foxwell, B.M.J., Feldmann, M., Mazi, M., 2010. Tumor necrosis factor- α triggers a cytokine cascade yielding postoperative cognitive decline. *Proc. Natl. Acad. Sci. USA* 107 (47), 20518–20522.
- Thompson, P.J., Shah, A., Ntranos, V., Van Gool, F., Atkinson, M., Bhushan, A., 2019. Targeted elimination of senescent beta cells prevents type 1 diabetes. *Cell Metab.* 29 (5).
- Tichauer, J.E., Flores, B., Soler, B., Eugenín-von Bernhardt, L., Ramírez, G., von Bernhardt, R., 2014. Age-dependent changes on TGF β 1 Smad3 pathway modify the pattern of microglial cell activation. *Brain Behav. Immun.* 37, 187–196.
- Triana-Martínez, F., Picallós-Rabina, P., Da Silva-Álvarez, S., Pietroluca, F., Llanos, S., Rodilla, V., Soprano, E., Pedrosa, P., Ferreira, A., Barradas, M., Hernández-González, F., Lalinde, M., Prats, N., Bernadó, C., González, P., Gómez, M., Ikonopoulou, M.P., Fernández-Marcos, P.J., García-Caballero, T., Del Pino, P., Arribas, J., Vidal, A., González-Barcia, M., Serrano, M., Loza, M.I., Domínguez, E., Collado, M., 2019. Identification and characterization of Cardiac Glycosides as senolytic compounds. *Nat. Commun.* 10 (1), 4731.
- Tse, C., Shoemaker, A.R., Adickes, J., Anderson, M.G., Chen, J., Jin, S., Johnson, E.F., Marsh, K.C., Mitten, M.J., Nimmer, P., Roberts, L., Tahir, S.K., Xiao, Y., Yang, X., Zhang, H., Fesik, S., Rosenberg, S.H., Elmore, S.W., 2008. ABT-263: a potent and orally bioavailable Bcl-2 family inhibitor. *Cancer Res* 68 (9), 3421–3428.
- Turnquist, C., Horikawa, I., Foran, E., Major, E.O., Vojtesek, B., Lane, D.P., Lu, X., Harris, B.T., Harris, C.C., 2016. p53 isoforms regulate astrocyte-mediated neuroprotection and neurodegeneration. *Cell Death Differ.* 23 (9), 1515–1528.
- Ungvari, Z., Tarantini, S., Donato, A.J., Galvan, V., Csizsar, A., 2018. Mechanisms of vascular aging. *Circ. Res* 123 (7), 849–867.
- Vaziri, H., Benchimol, S., 1998. Reconstitution of telomerase activity in normal human cells leads to elongation of telomeres and extended replicative life span. *Curr. Biol.* 8 (5), 279–282.
- Wang, B., Han, J., Elisseeff, J.H., Demaria, M., 2024. The senescence-associated secretory phenotype and its physiological and pathological implications. *Nat. Rev. Mol. Cell Biol.* 25 (12), 958–978.
- Wang, Q., Liu, T., Fang, Y., Xie, S., Huang, X., Mahmood, R., Ramaswamy, G., Sakamoto, K.M., Darzynkiewicz, Z., Xu, M., Dai, W., 2003. BUBR1 deficiency results in abnormal megakaryopoiesis. *Blood* 103 (4), 1278–1285.
- Wang, R., Yu, Z., Sunchu, B., Shoaf, J., Dang, I., Zhao, S., Caples, K., Bradley, L., Beaver, L.M., Ho, E., Löhr, C.V., Perez, V.I., 2017. Rapamycin inhibits the secretory phenotype of senescent cells by a Nr2f2-independent mechanism. *Aging Cell* 16 (3), 564–574.
- Wang, W., Zhao, F., Ma, X., Perry, G., Zhu, X., 2020. Mitochondria dysfunction in the pathogenesis of Alzheimer's disease: recent advances. *Mol. Neurodegener.* 15 (1), 30.
- Wang, Y., Chang, J., Liu, X., Zhang, X., Zhang, S., Zhang, X., Zhou, D., Zheng, G., 2016. Discovery of piperlongumine as a potential novel lead for the development of senolytic agents. *Aging (Albany NY)* 8 (11), 2915–2926.
- Washington University School of Medicine., 2021. Senolytic Therapy to Modulate the Progression of Alzheimer's Disease (StoMP-AD) Study (StoMP-AD), (<https://clinicaltrials.gov/study/NCT04685590#study-record-dates>).
- Weichhart, T., 2018. mTOR as regulator of lifespan, aging, and cellular senescence: a mini-review. *Gerontology* 64 (2), 127–134.
- Whyte, L.S., Lau, A.A., Hemsley, K.M., Hopwood, J.J., Sargeant, T.J., 2017. Endo-lysosomal and autophagic dysfunction: a driving factor in Alzheimer's disease? *J. Neurochem.* 140 (5), 703–717.
- Wu, W., Fu, J., Gu, Y., Wei, Y., Ma, P., Wu, J., 2020. JAK2/STAT3 regulates estrogen-related senescence of bone marrow stem cells. *J. Endocrinol.* 245 (1), 141–153.
- Xu, L., Botchway, B.O.A., Zhang, S., Zhou, J., Liu, X., 2018. Inhibition of NF- κ B signaling pathway by resveratrol improves spinal cord injury. *Front. Neurosci.* 12, 690.
- Xu, M., Tchkonina, T., Ding, H., Ogronnik, M., Lubbers, E.R., Pirtskhalava, T., White, T.A., Johnson, K.O., Stout, M.B., Mezera, V., Giorgadze, N., Jensen, M.D., LeBrasseur, N. K., Kirkland, J.L., 2015b. JAK inhibition alleviates the cellular senescence-associated secretory phenotype and frailty in old age. *Proc. Natl. Acad. Sci. USA* 112 (46), E6301–E6310.
- Xu, M., Palmer, A.K., Ding, H., Weivoda, M.M., Pirtskhalava, T., White, T.A., Sepe, A., Johnson, K.O., Stout, M.B., Giorgadze, N., Jensen, M.D., LeBrasseur, N.K., Tchkonina, T., Kirkland, J.L., 2015a. Targeting senescent cells enhances adipogenesis and metabolic function in old age. *Elife* 4, e12997.
- Xu, T., Sun, D., Chen, Y., Ouyang, L., 2020. Targeting mTOR for fighting diseases: a revisited review of mTOR inhibitors. *Eur. J. Med. Chem.* 199, 112391.
- Yamazaki, Y., Baker, D.J., Tachibana, M., Liu, C.-C., van Deursen, J.M., Brott, T.G., Bu, G., Kanekiyo, T., 2016. Vascular cell senescence contributes to blood-brain barrier breakdown. *Stroke* 47 (4), 1068–1077.
- Yang, Z., Jun, H., Choi, C.-I., Yoo, K.H., Cho, C.H., Hussaini, S.M.Q., Simmons, A.J., Kim, S., van Deursen, J.M., Baker, D.J., Jang, M.-H., 2017. Age-related decline in BubR1 impairs adult hippocampal neurogenesis. *Aging Cell* 16 (3), 598–601.
- Yosef, R., Pilpel, N., Tokarsky-Amiel, R., Biran, A., Ovadya, Y., Cohen, S., Vadai, E., Dassa, L., Shahar, E., Condiotti, R., Ben-Porath, I., Krizhanovsky, V., 2016. Directed elimination of senescent cells by inhibition of BCL-W and BCL-XL. *Nat. Commun.* 7, 11190.
- Yousefzadeh, M.J., Zhu, Y., McGowan, S.J., Angelini, L., Fuhrmann-Stroissnig, H., Xu, M., Ling, Y.Y., Melos, K.I., Pirtskhalava, T., Inman, C.L., McGuckian, C., Wade, E. A., Kato, J.I., Grassi, D., Wentworth, M., Burd, C.E., Arriaga, E.A., Ladiges, W.L., Tchkonina, T., Kirkland, J.L., Robbins, P.D., Niedernhofer, L.J., 2018. Fisetin is a senotherapeutic that extends health and lifespan. *EBioMedicine* 36, 18–28.
- Yu, H.-M., Zhao, Y.-M., Luo, X.-G., Feng, Y., Ren, Y., Shang, H., He, Z.-Y., Luo, X.-M., Chen, S.-D., Wang, X.-Y., 2012. Repeated lipopolysaccharide stimulation induces cellular senescence in BV2 cells. *Neuroimmunomodulation* 19 (2), 131–136.
- Zhang, C., Xie, Y., Chen, H., Lv, L., Yao, J., Zhang, M., Xia, K., Feng, X., Li, Y., Liang, X., Sun, X., Deng, C., Liu, G., 2020. FOXO4-DRI alleviates age-related testosterone secretion insufficiency by targeting senescent Leydig cells in aged mice. *Aging (Albany NY)* 12 (2), 1272–1284.
- Zhang, L., Zhao, J., Mu, X., McGowan, S.J., Angelini, L., O'Kelly, R.D., Yousefzadeh, M.J., Sakamoto, A., Aversa, Z., LeBrasseur, N.K., Suh, Y., Huard, J., Kamenecka, T.M., Niedernhofer, L.J., Robbins, P.D., 2021. Novel small molecule inhibition of IKK/NF- κ B activation reduces markers of senescence and improves healthspan in mouse models of aging. *Aging Cell* 20 (12), e13486.
- Zhang, P., Kishimoto, Y., Grammatikakis, I., Gottmukkala, K., Cutler, R.G., Zhang, S., Abdelmohsen, K., Bohr, V.A., Misra Sen, J., Gorospe, M., Mattson, M.P., 2019. Senolytic therapy alleviates A β -associated oligodendrocyte progenitor cell senescence and cognitive deficits in an Alzheimer's disease model. *Nat. Neurosci.* 22 (5), 719–728.
- Zhang, X., Zhang, S., Liu, X., Wang, Y., Chang, J., Zhang, X., Mackintosh, S.G., Tackett, A. J., He, Y., Lv, D., Laberge, R.-M., Campisi, J., Wang, J., Zheng, G., Zhou, D., 2018. Oxidation resistance 1 is a novel senolytic target. *Aging Cell* 17 (4), e12780.
- Zhao, J., Zhang, L., Mu, X., Doebelin, C., Nguyen, W., Wallace, C., Reay, D.P., McGowan, S.J., Corbo, L., Clemens, P.R., Wilson, G.M., Watkins, S.C., Solt, L.A., Cameron, M.D., Huard, J., Niedernhofer, L.J., Kamenecka, T.M., Robbins, P.D., 2018. Development of novel NEMO-binding domain mimetics for inhibiting IKK/NF- κ B activation. *PLoS Biol.* 16 (6), e2004663.
- Zhao, J., Zhang, L., Lu, A., Han, Y., Colangelo, D., Bukata, C., Scibetta, A., Yousefzadeh, M.J., Li, X., Gurkar, A.U., McGowan, S.J., Angelini, L., O'Kelly, R., Li, H., Corbo, L., Sano, T., Nick, H., Pola, E., Pilla, S.P.S., Ladiges, W.C., Vo, N., Huard, J., Niedernhofer, L.J., Robbins, P.D., 2020. ATM is a key driver of NF- κ B-dependent DNA-damage-induced senescence, stem cell dysfunction and aging. *Aging (Albany NY)* 12 (6), 4688–4710.
- Zheng, Y., Yang, J., Li, X., Qi, L., Zheng, Z., Kong, J., Zhang, G., Guo, Y., 2025. Mitochondria at the crossroads: Quality control mechanisms in neuronal senescence and neurodegeneration. *Neurobiol. Dis.* 208, 106862.
- Zhu, Y., Tchkonina, T., Pirtskhalava, T., Gower, A.C., Ding, H., Giorgadze, N., Palmer, A. K., Ikono, Y., Hubbard, G.B., Lenburg, M., O'Hara, S.P., LaRusso, N.F., Miller, J.D., Roos, C.M., Verzosa, G.C., LeBrasseur, N.K., Wren, J.D., Farr, J.N., Khosla, S., Stout, M.B., McGowan, S.J., Fuhrmann-Stroissnig, H., Gurkar, A.U., Zhao, J., Colangelo, D., Dorronsoro, A., Ling, Y.Y., Barghouthy, A.S., Navarro, D.C., Sano, T.,

- Robbins, P.D., Niedernhofer, L.J., Kirkland, J.L., 2015. The Achilles' heel of senescent cells: from transcriptome to senolytic drugs. *Aging Cell* 14 (4), 644–658.
- Zhu, Y., Doornebal, E.J., Pirtskhalava, T., Giorgadze, N., Wentworth, M., Fuhrmann-Stroissnigg, H., Niedernhofer, L.J., Robbins, P.D., Tchkonja, T., Kirkland, J.L., 2017. New agents that target senescent cells: the flavone, fisetin, and the BCL-XL inhibitors, A1331852 and A1155463. *Aging (Albany NY)* 9 (3), 955–963.
- Zou, Y., Zhang, N., Ellerby, L.M., Davalos, A.R., Zeng, X., Campisi, J., Desprez, P.-Y., 2012. Responses of human embryonic stem cells and their differentiated progeny to ionizing radiation. *Biochem Biophys. Res Commun.* 426 (1), 100–105.